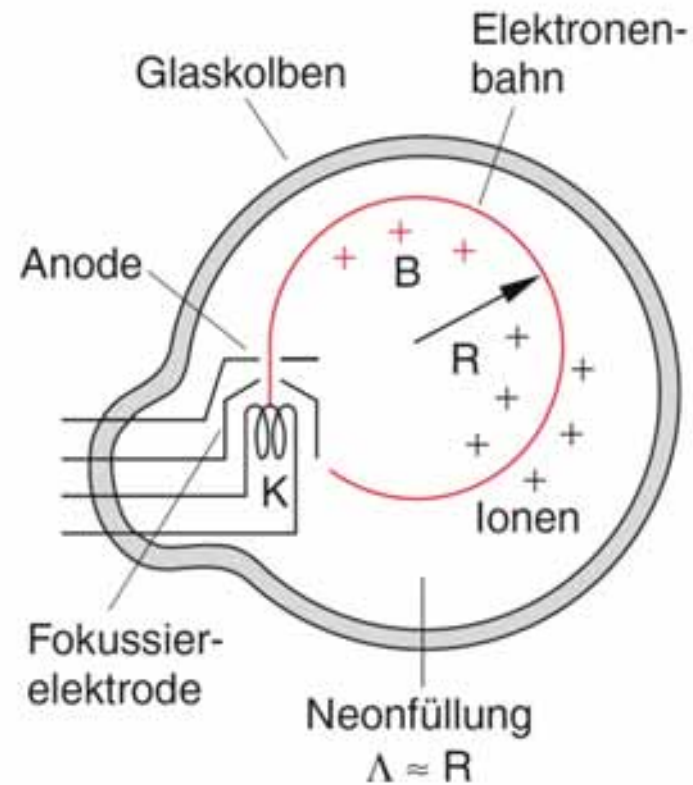


# Elektrizität und Magnetismus

## Kapitel 4: Magnetfelder

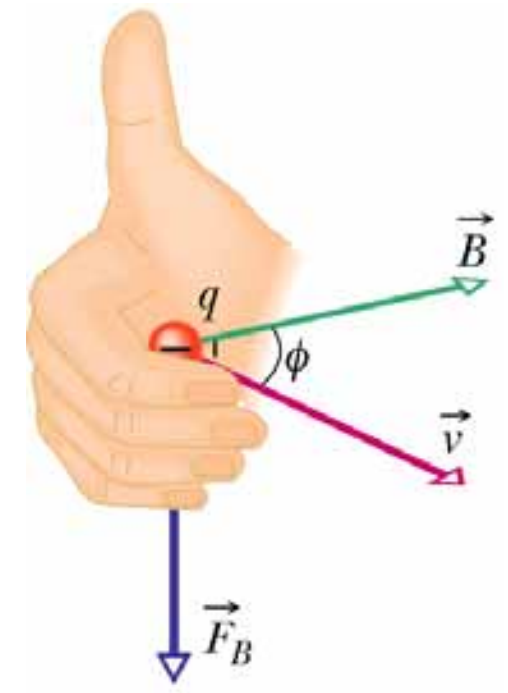
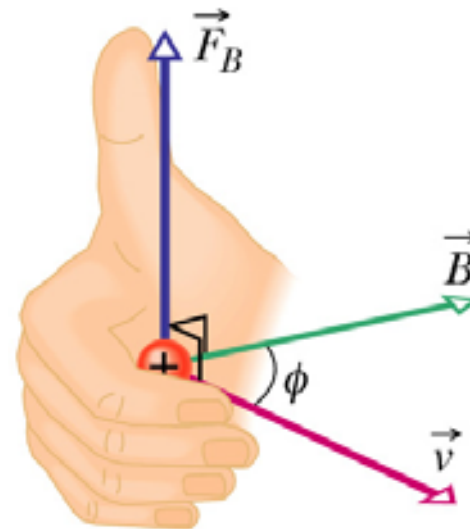
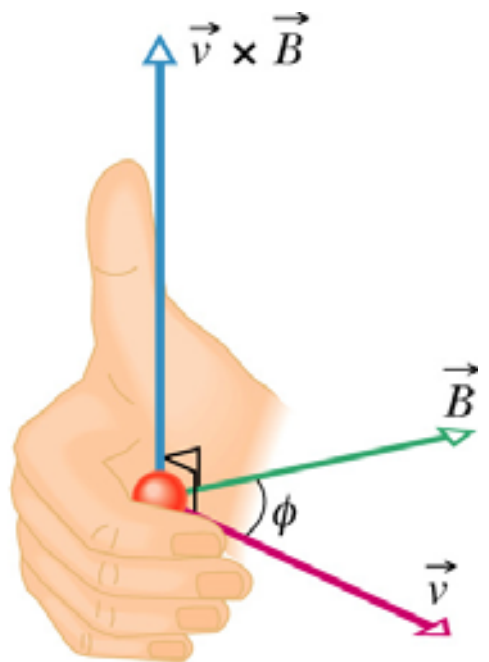
Prof. Axel Görlitz  
Heinrich-Heine Universität Düsseldorf  
[Axel.Goerlitz@uni-duesseldorf.de](mailto:Axel.Goerlitz@uni-duesseldorf.de)

## zu 4.1: Fadenstrahlrohr



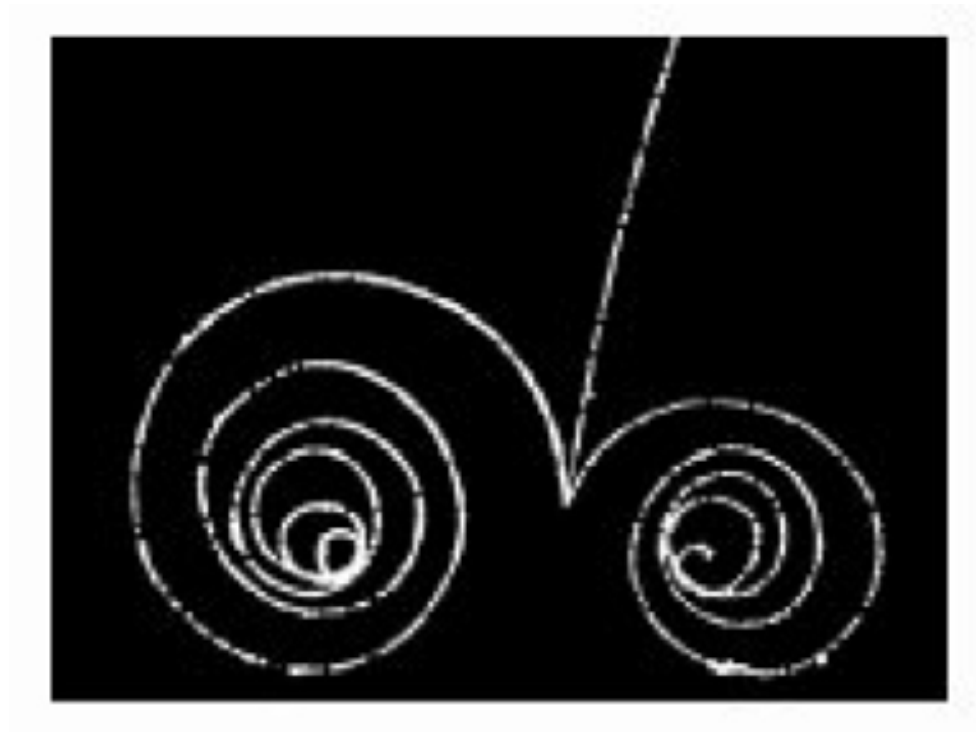
Demtröder – Experimentalphysik 2 (2013), Abb. 3.24

## zu 4.1: Rechte-Hand-Regel



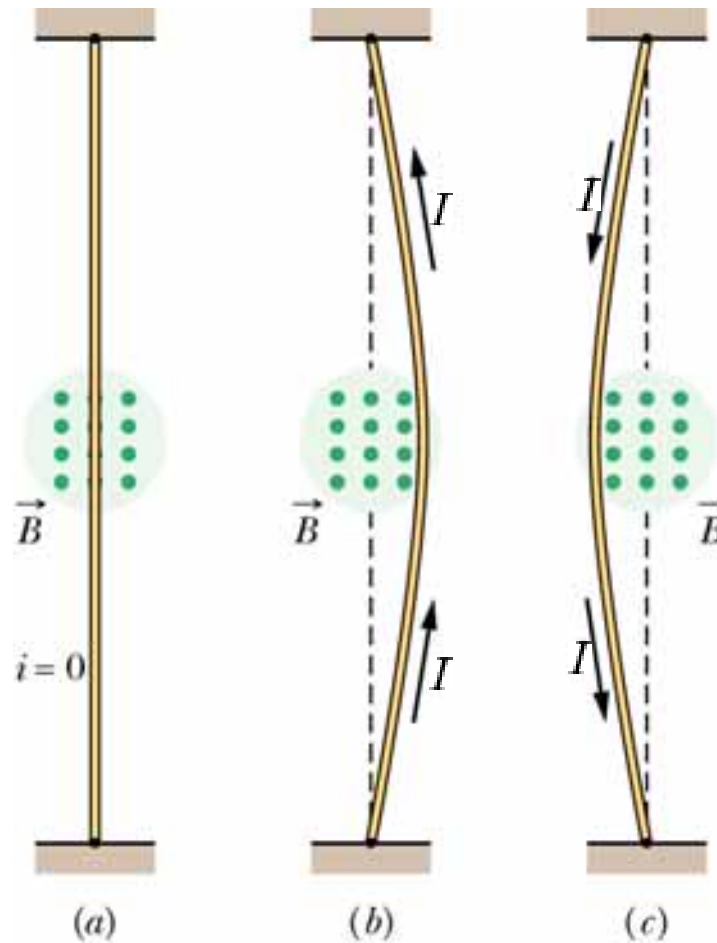
Halliday-Resnick-Walker (2003) Abb. 29.2

## zu 4.1: Elektron und Positron im Magnetfeld



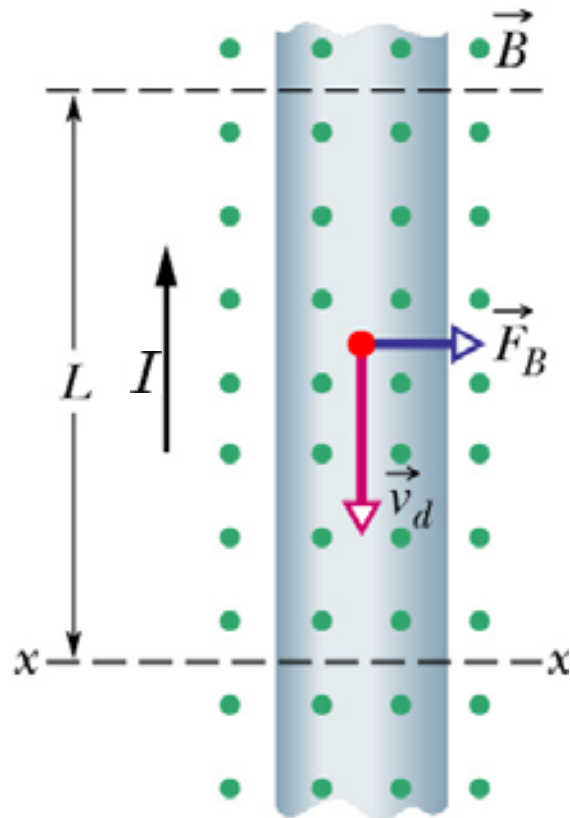
<https://www.spektrum.de/lexikon/physik/vakuum/15034>

## zu 4.2: Gerader Leiter im Magnetfeld

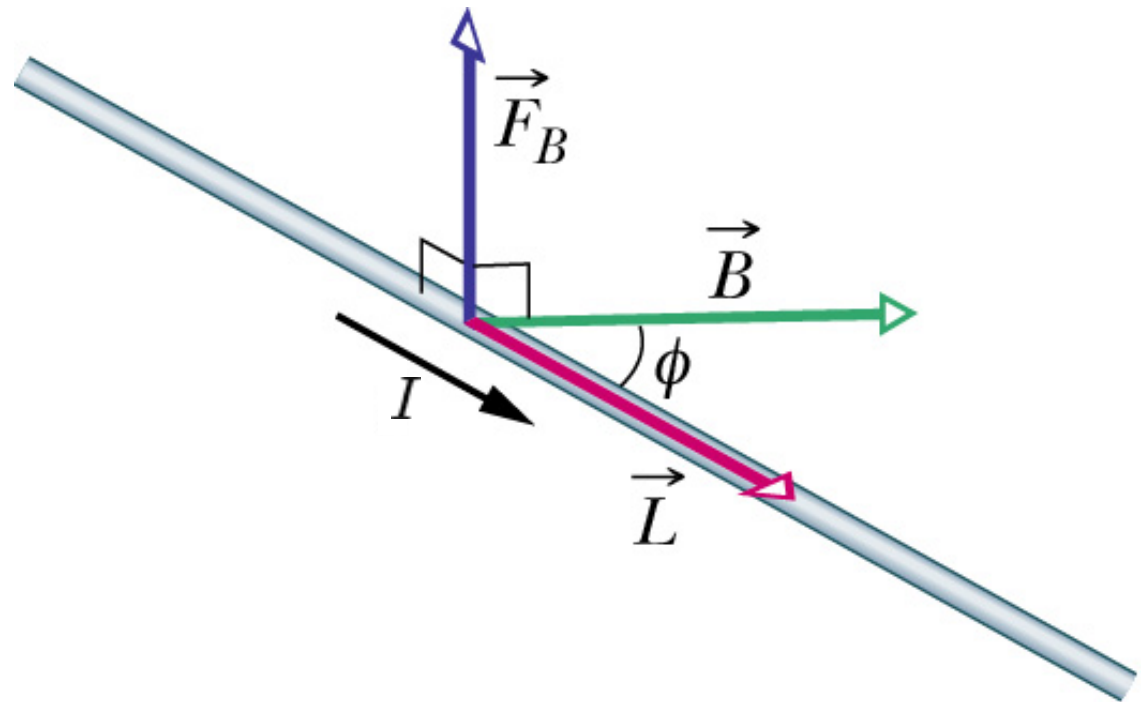


Halliday-Resnick-Walker (2003) Abb. 29.16

## zu 4.2: Gerader Leiter im Magnetfeld

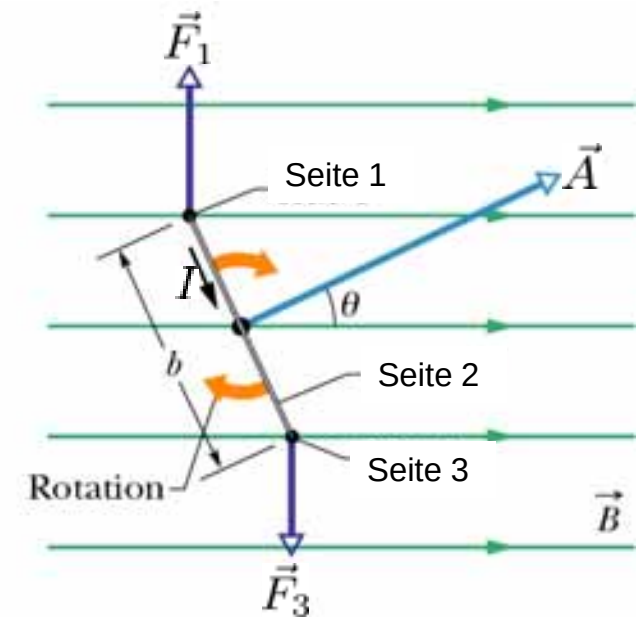
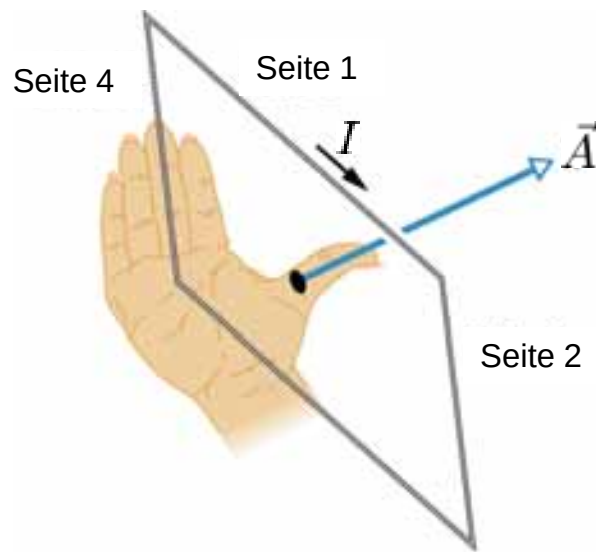
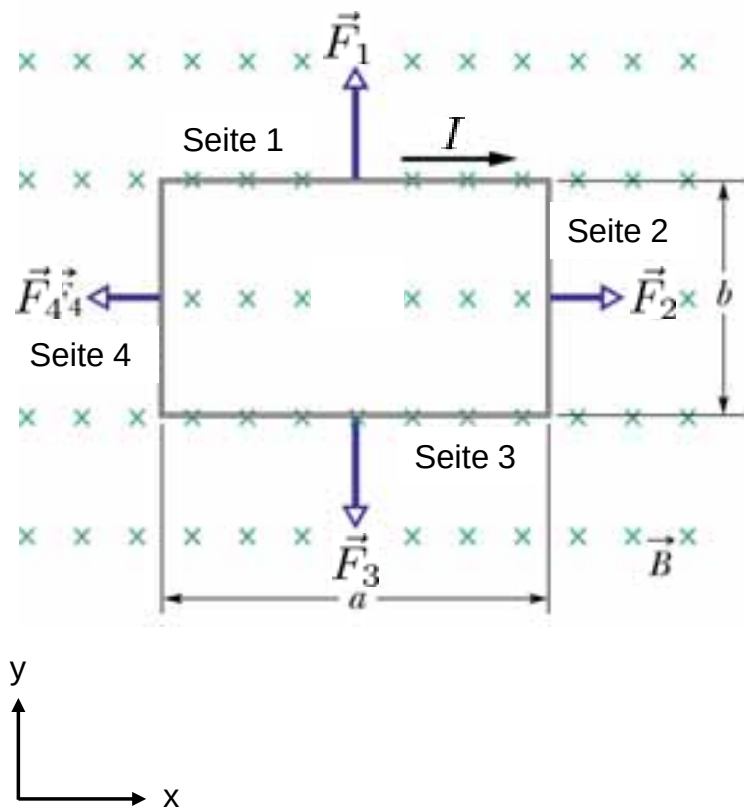


Halliday-Resnick-Walker (2003) Abb. 29.17



Halliday-Resnick-Walker (2003) Abb. 29.17

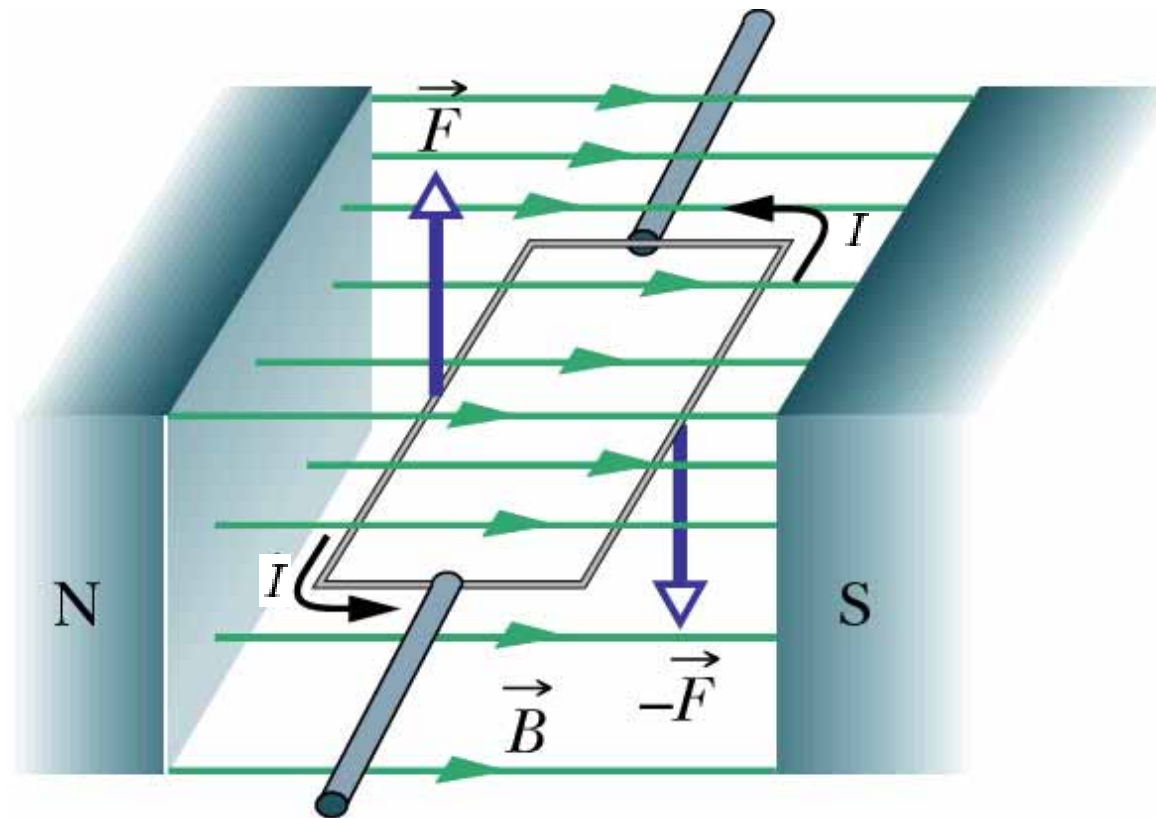
## zu 4.2: Leiterschleife im Magnetfeld



Halliday-Resnick-Walker (2003) Abb. 29.21

## zu 4.2: Leiterschleife im Magnetfeld

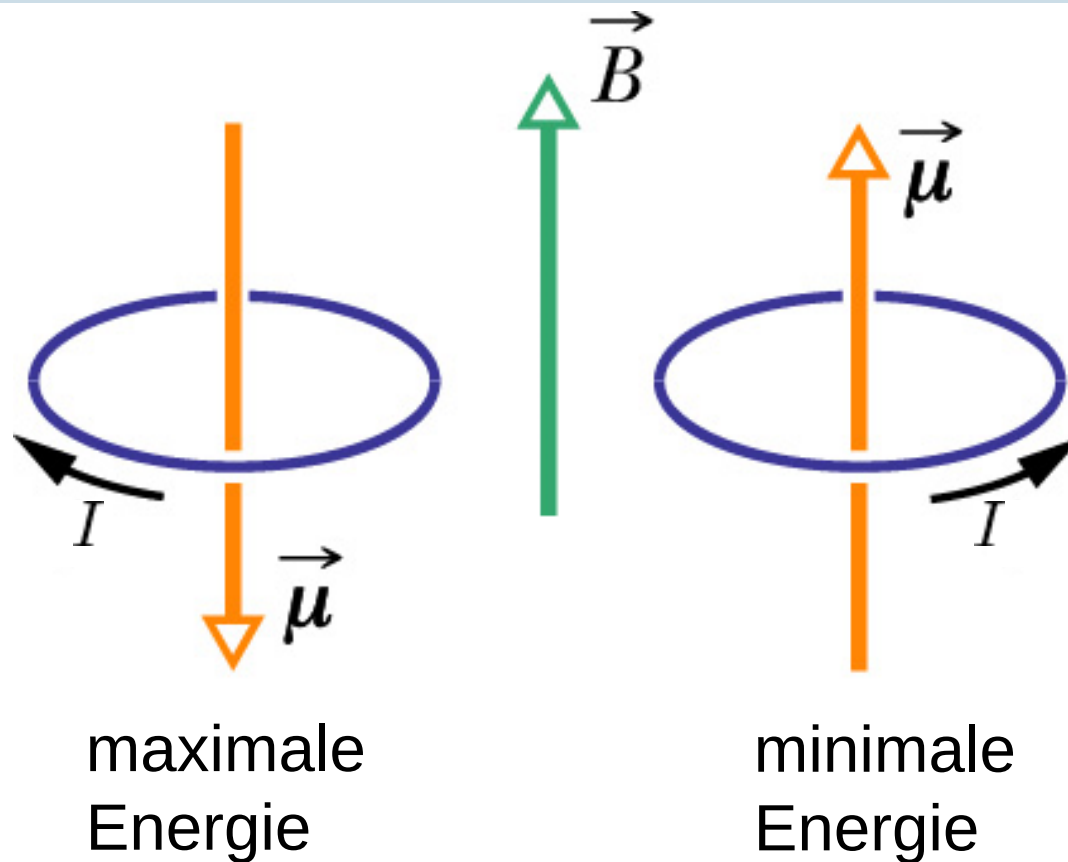
Prinzip eines  
Elektromotors



Halliday-Resnick-Walker (2003) Abb. 29.20



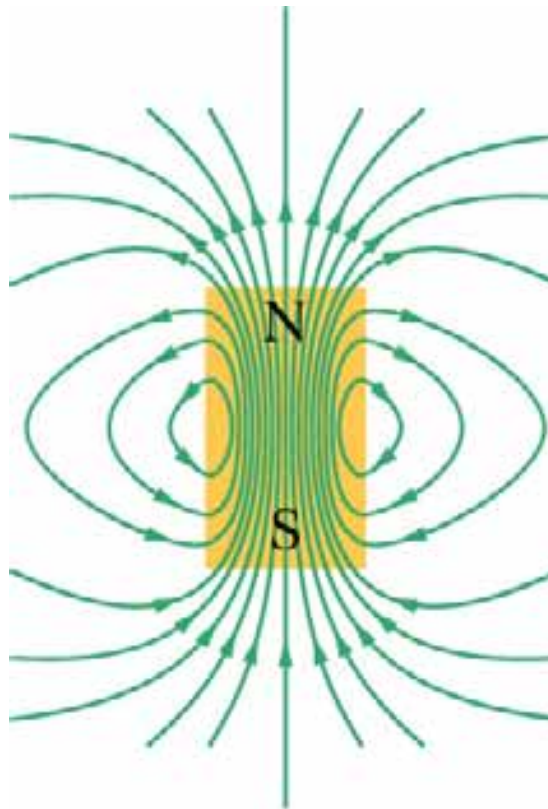
## zu 4.2: Magnetischer Dipol



Halliday-Resnick-Walker (2003) Abb. 29.23

## zu 4.3: Magnetische Feldlinien

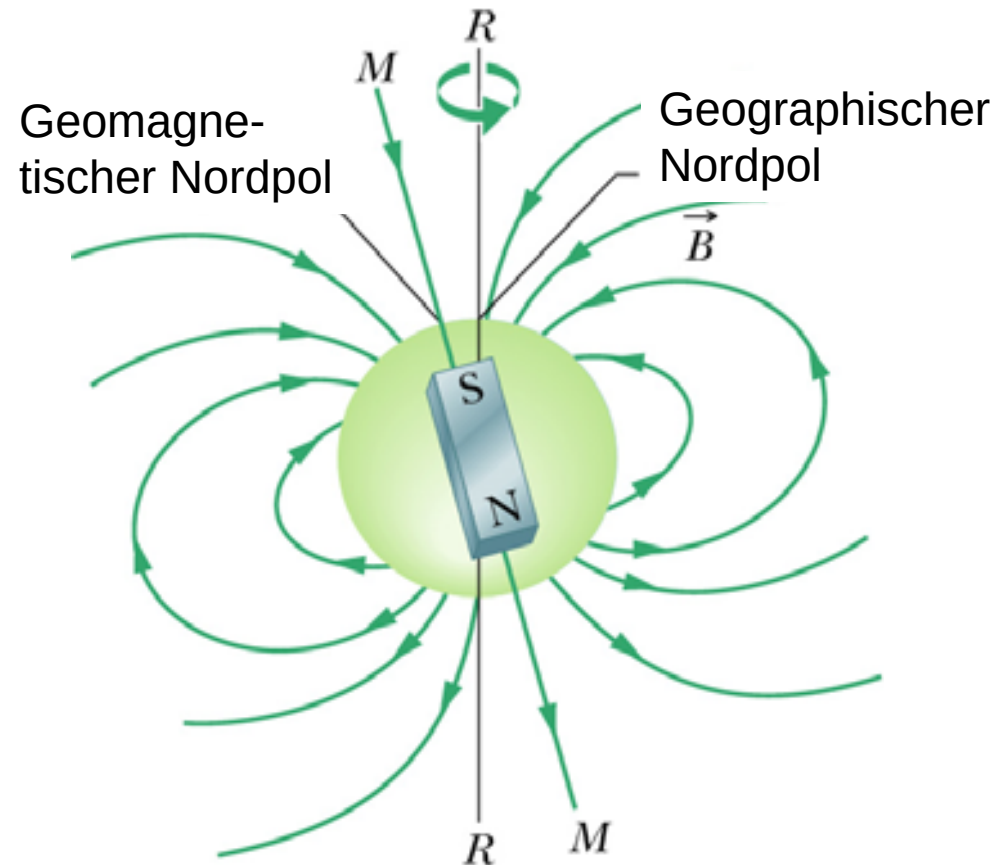
### Stabmagnet



Halliday-Resnick-Walker (2003) Abb. 29.04

## zu 4.3: Magnetische Feldlinien

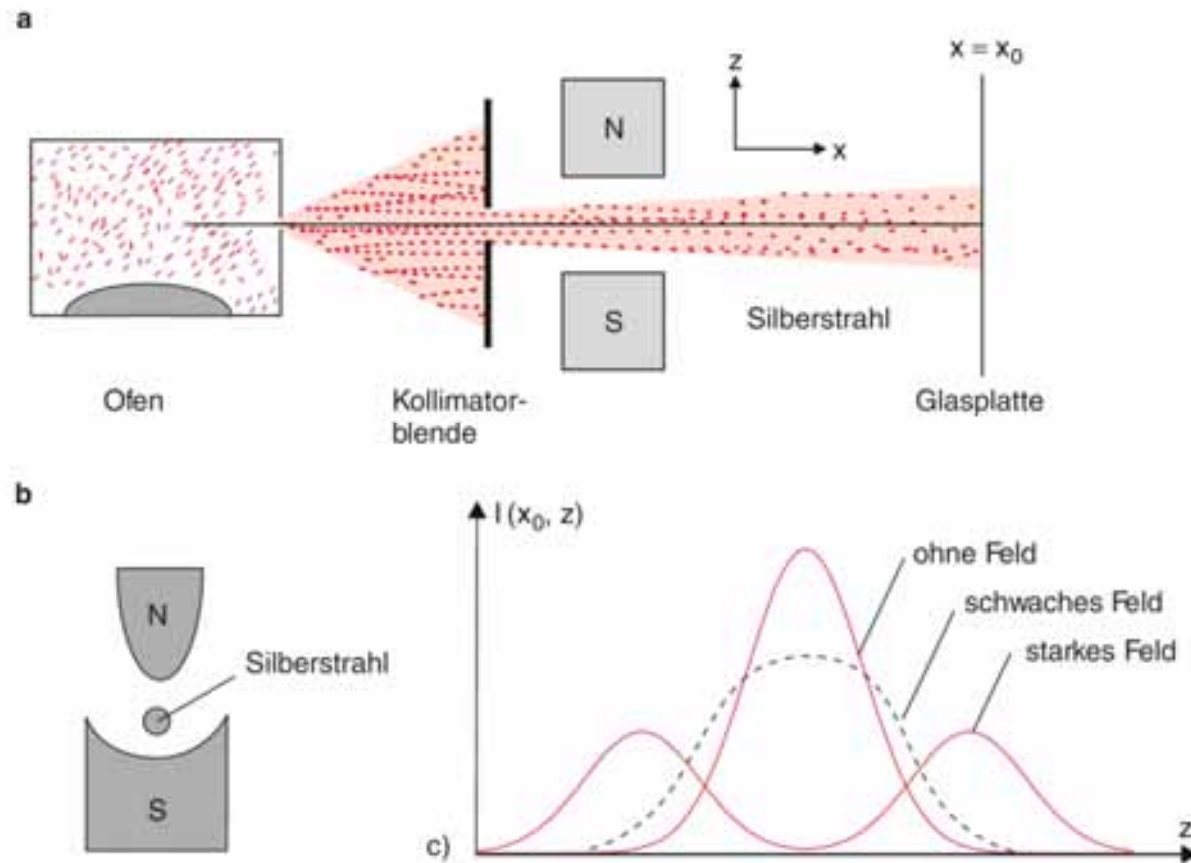
### Erdmagnetfeld



Halliday-Resnick-Walker (2003) Abb. 32.4

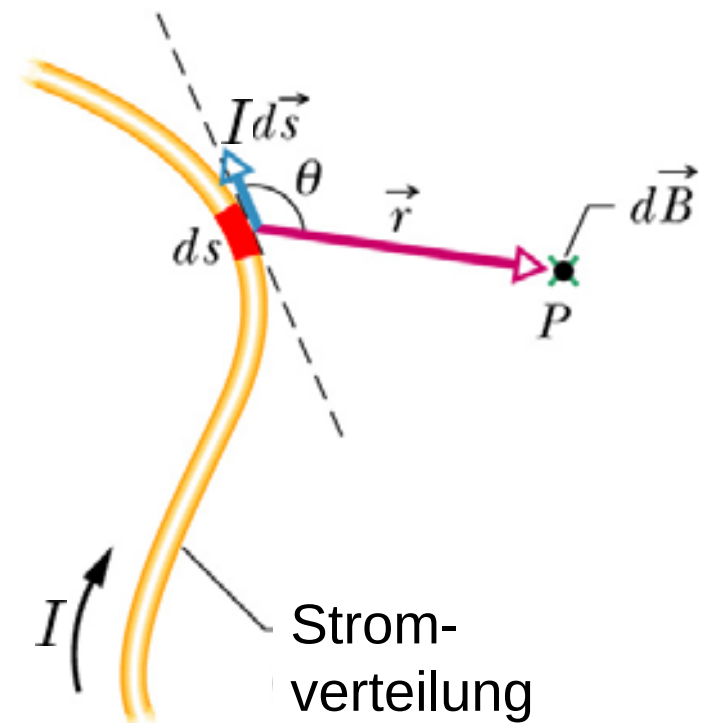
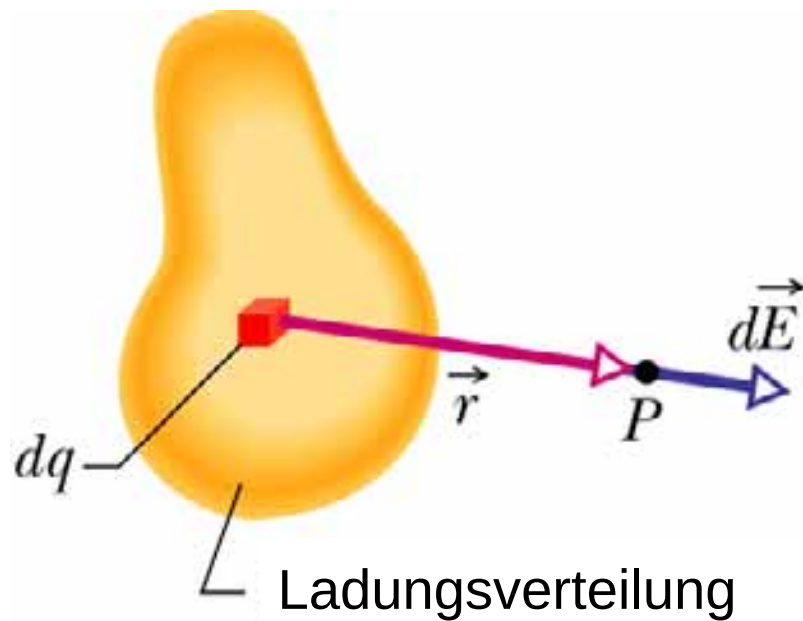
# zu 4.3: Kraft auf einen magnetischen Dipol

## Stern-Gerlach-Versuch



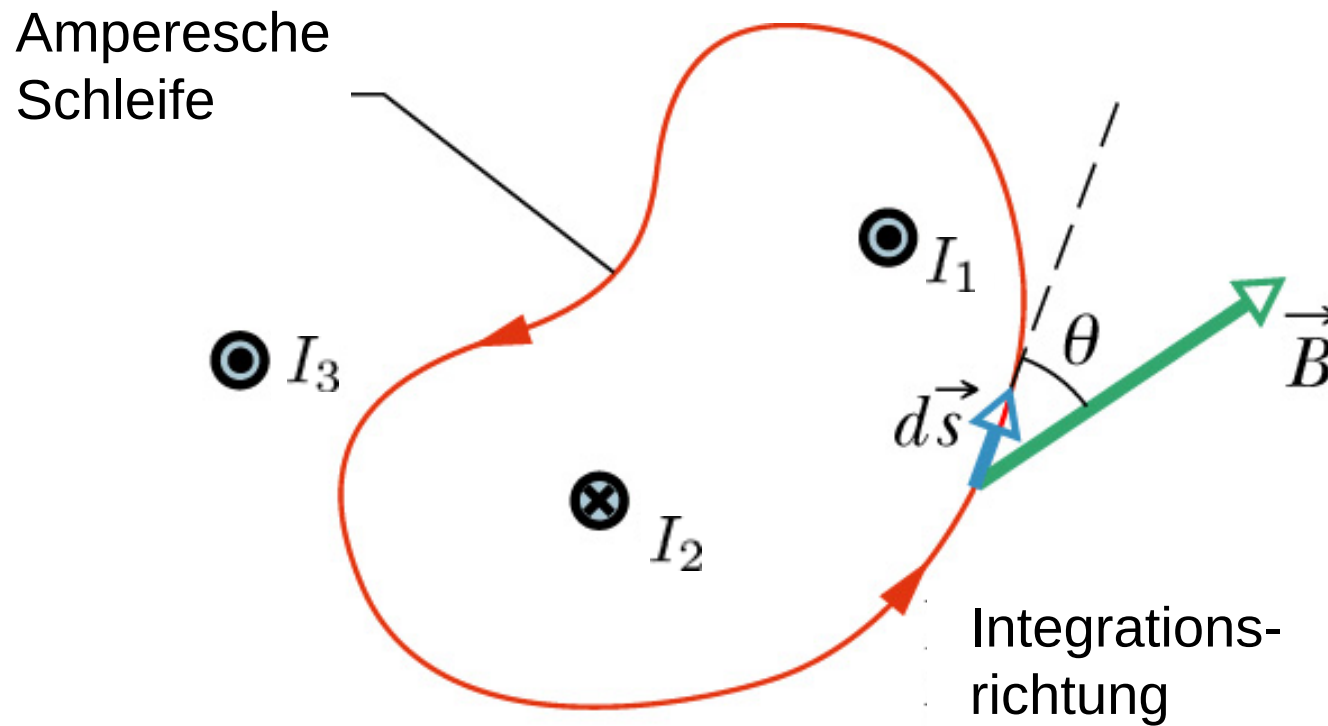
Demtröder – Experimentalphysik 3 (2013), Abb. 5.16

## zu 4.4: Gesetz von Biot-Savart



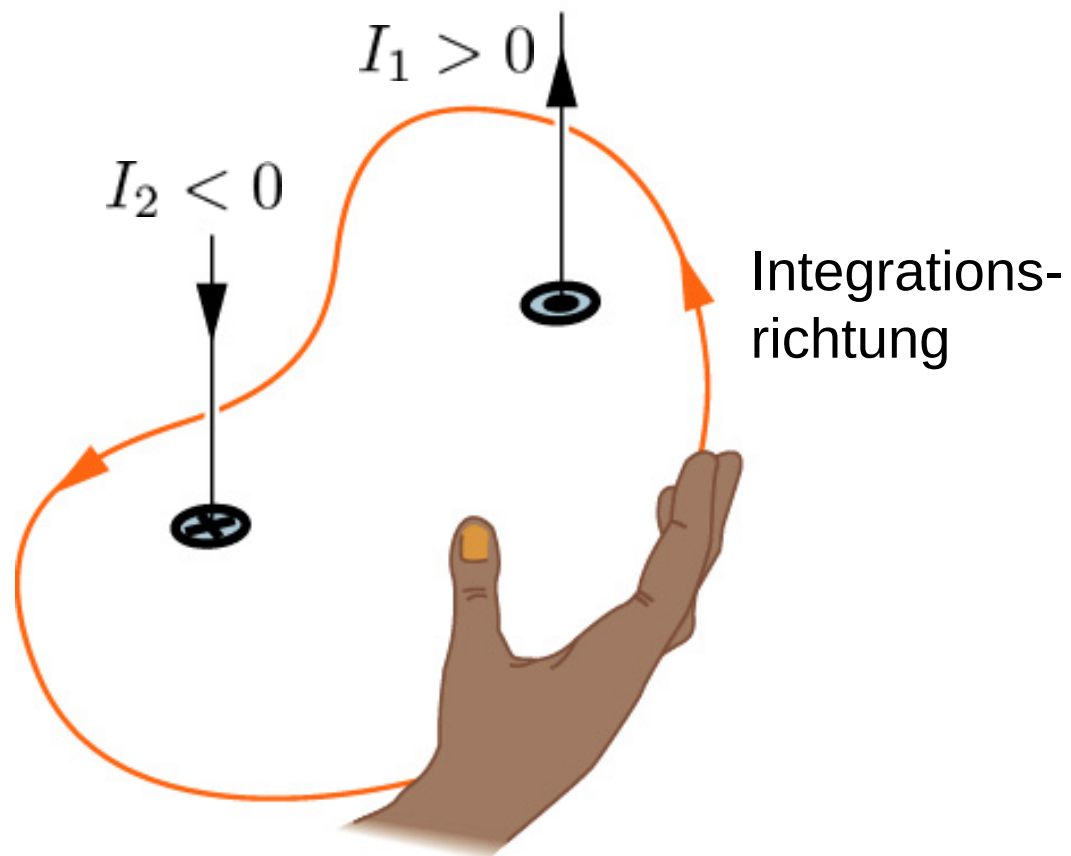
Halliday-Resnick-Walker (2003) Abb. 30.1

## zu 4.4: Amperesches Gesetz



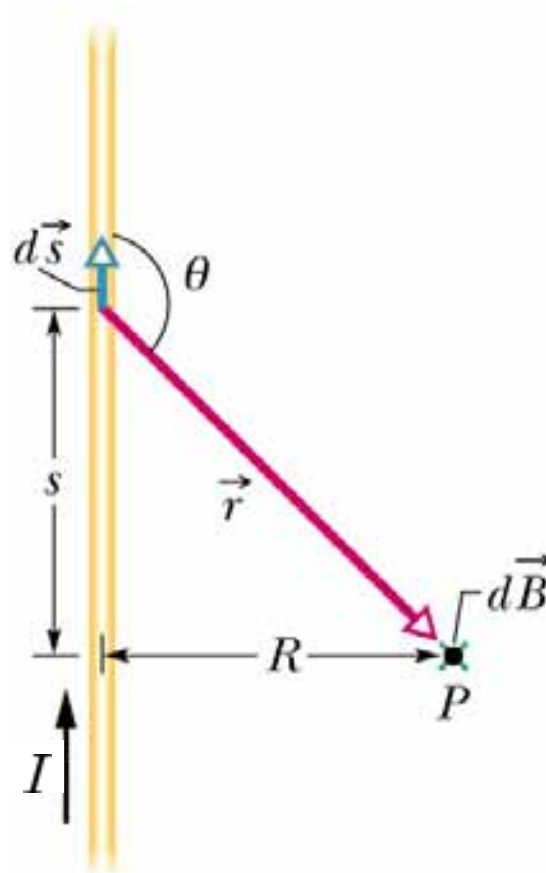
Halliday-Resnick-Walker (2003) Abb. 30.11

## zu 4.4: Amperesches Gesetz



Halliday-Resnick-Walker (2003) Abb. 30.11

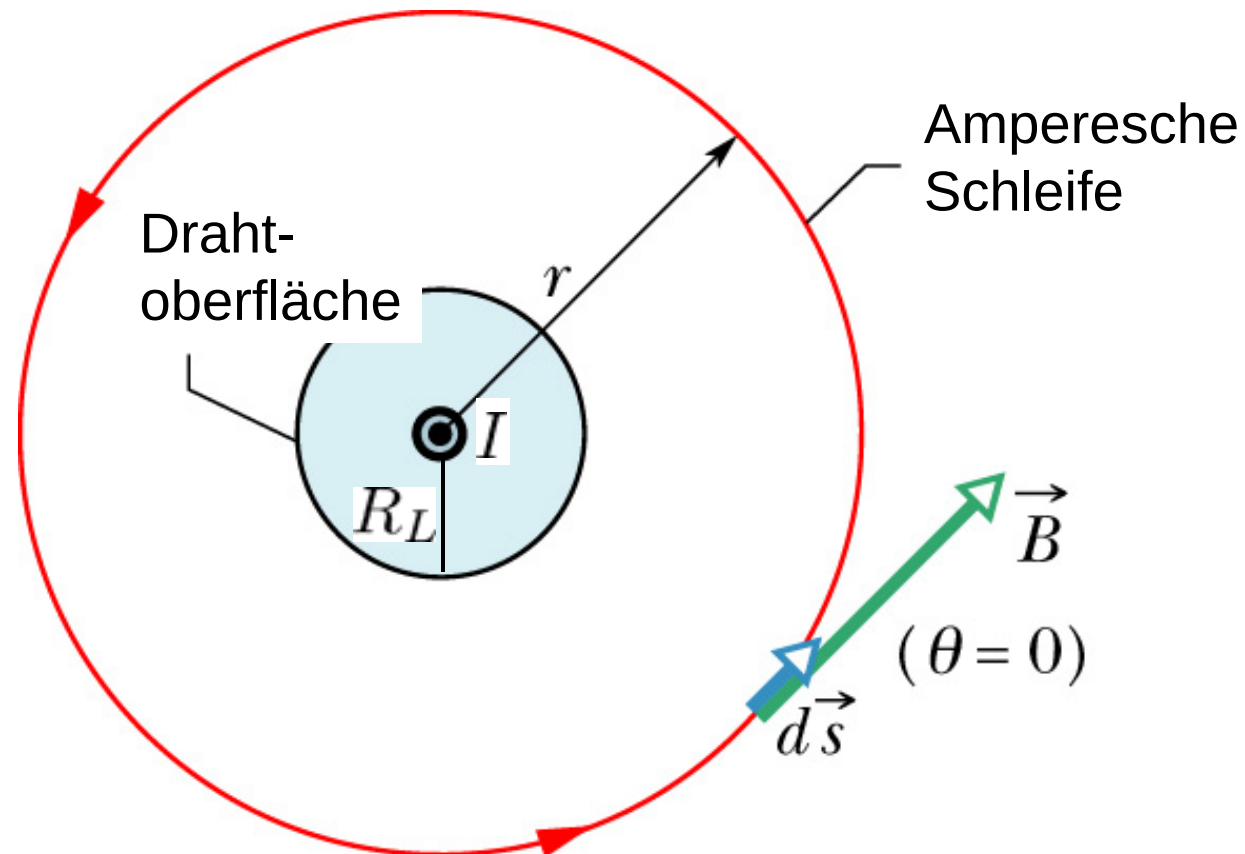
## zu 4.4: Magnetfeld eines geraden Leiters



Halliday-Resnick-Walker (2003) Abb. 30.5

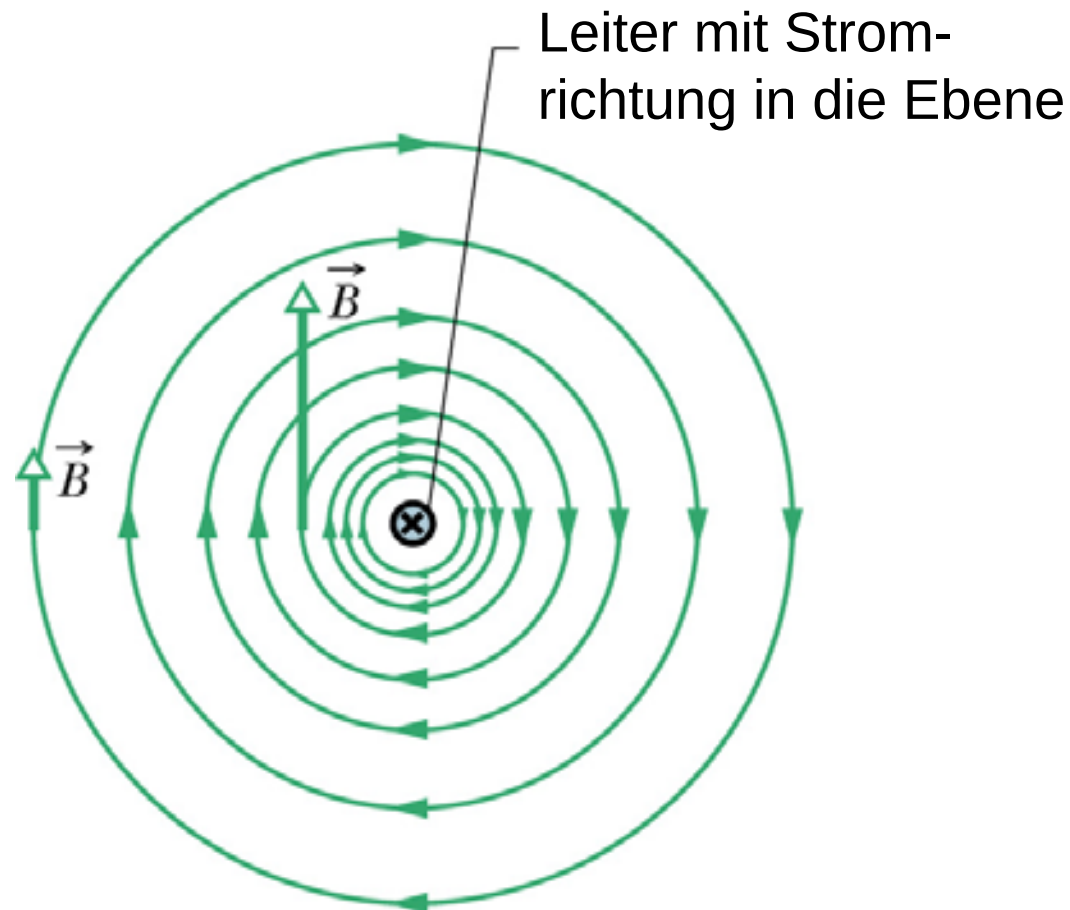


## zu 4.4: Magnetfeld eines geraden Leiters



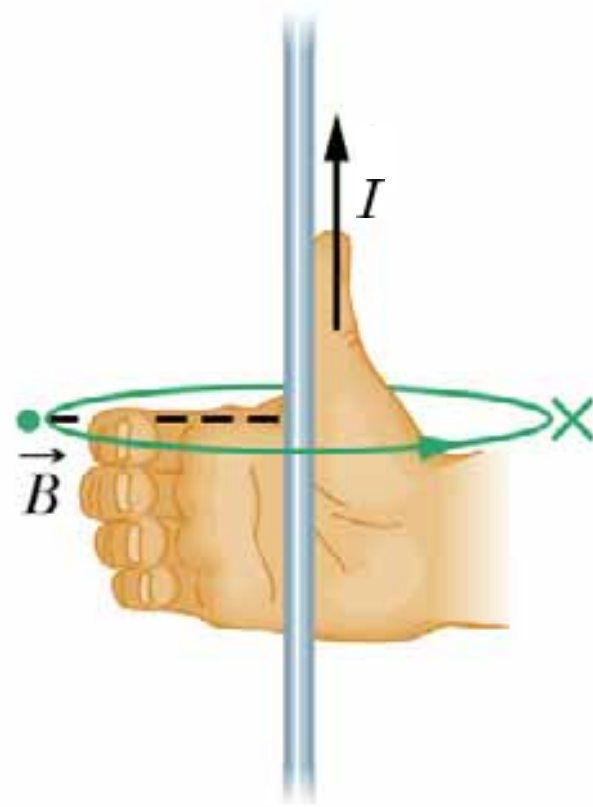
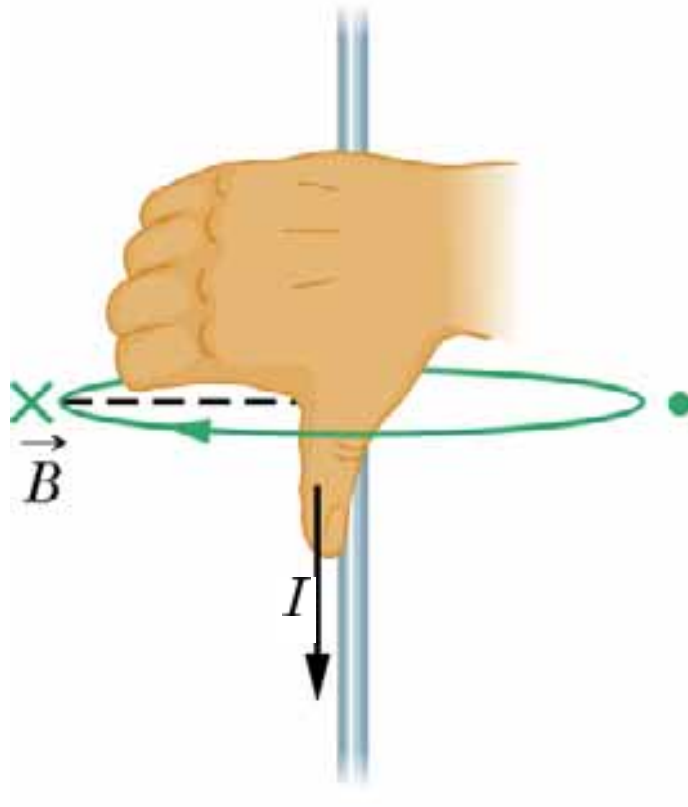
Halliday-Resnick-Walker (2003) Abb. 30.13

## zu 4.4: Magnetfeld eines geraden Leiters



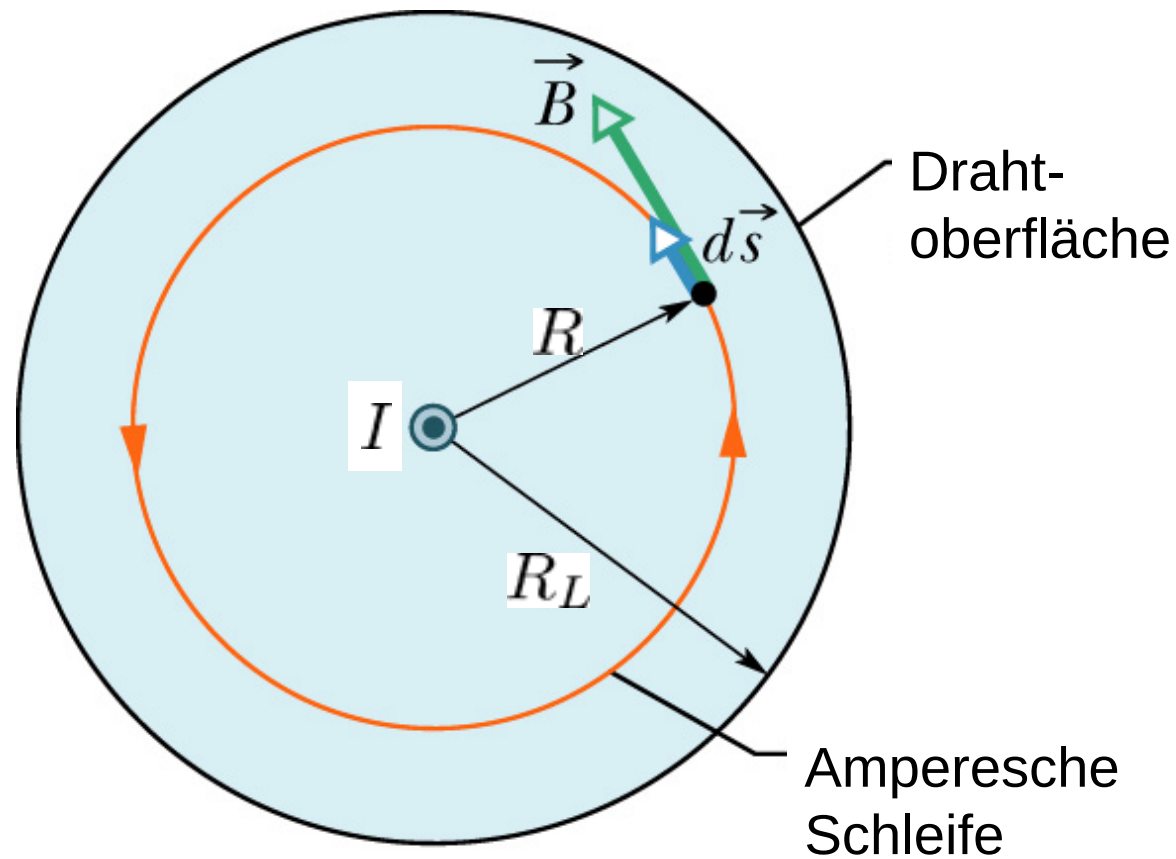
Halliday-Resnick-Walker (2003) Abb. 30.2

## zu 4.4: Rechte-Hand-Regel



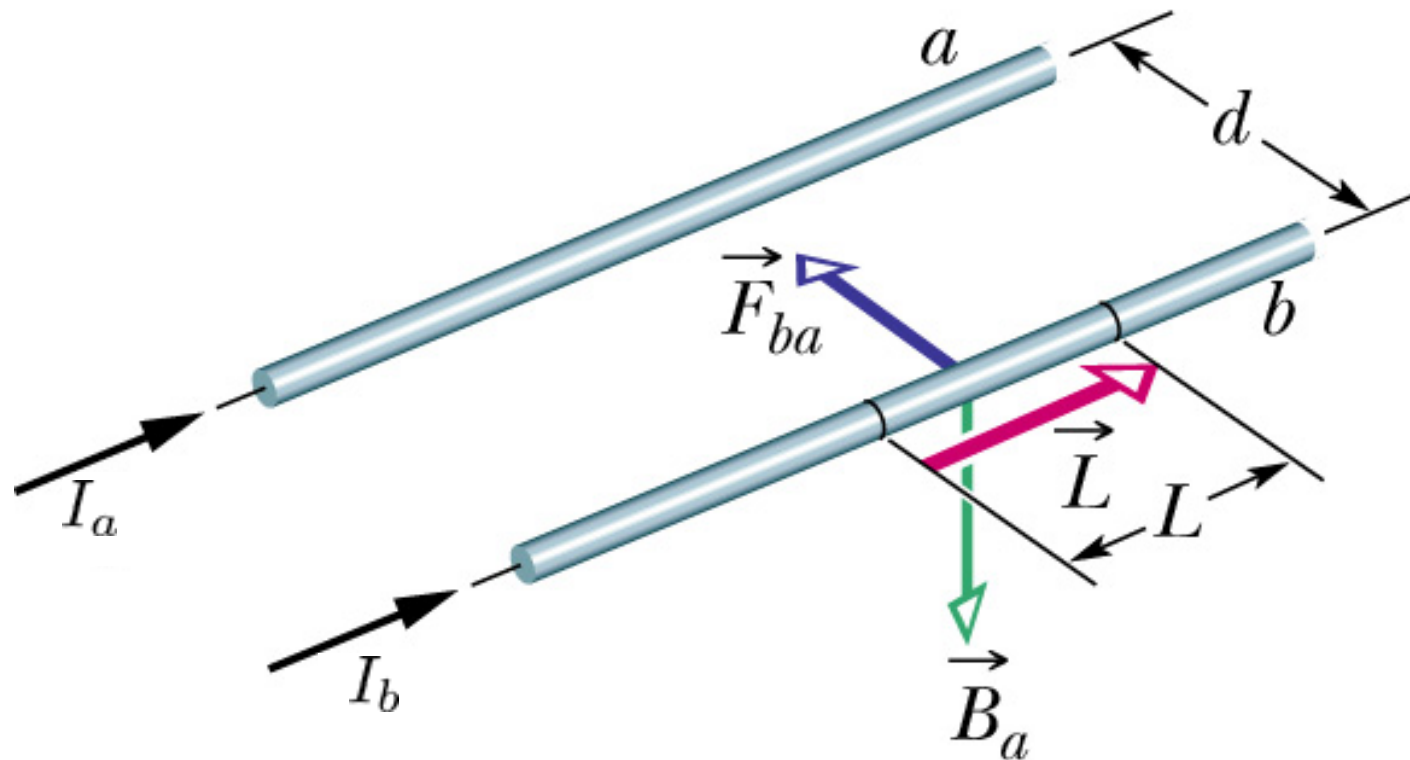
Halliday-Resnick-Walker (2003) Abb. 30.4

## zu 4.4: Magnetfeld eines geraden Leiters



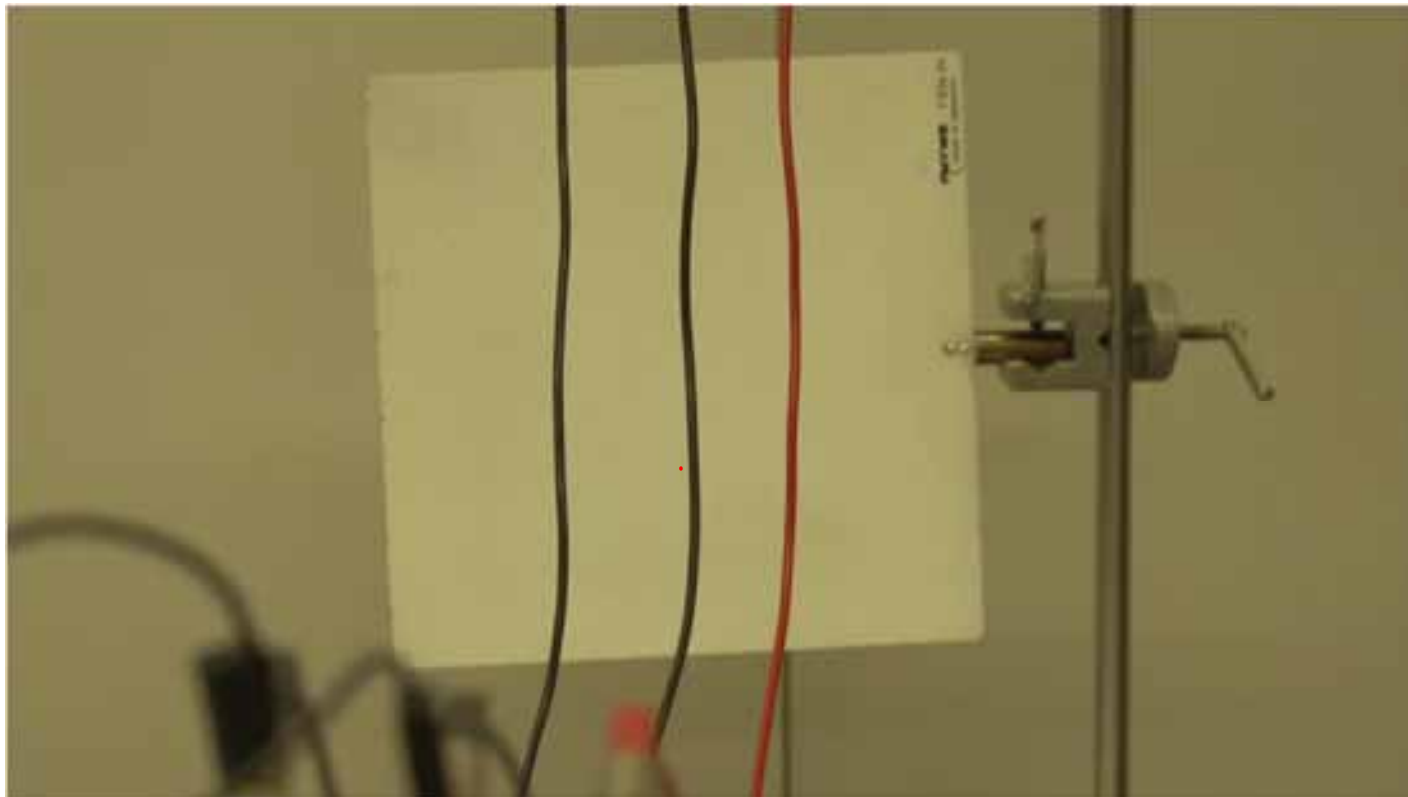
Halliday-Resnick-Walker (2003) Abb. 30.14

## zu 4.4: Kraft zwischen stromdurchflossenen Leitern

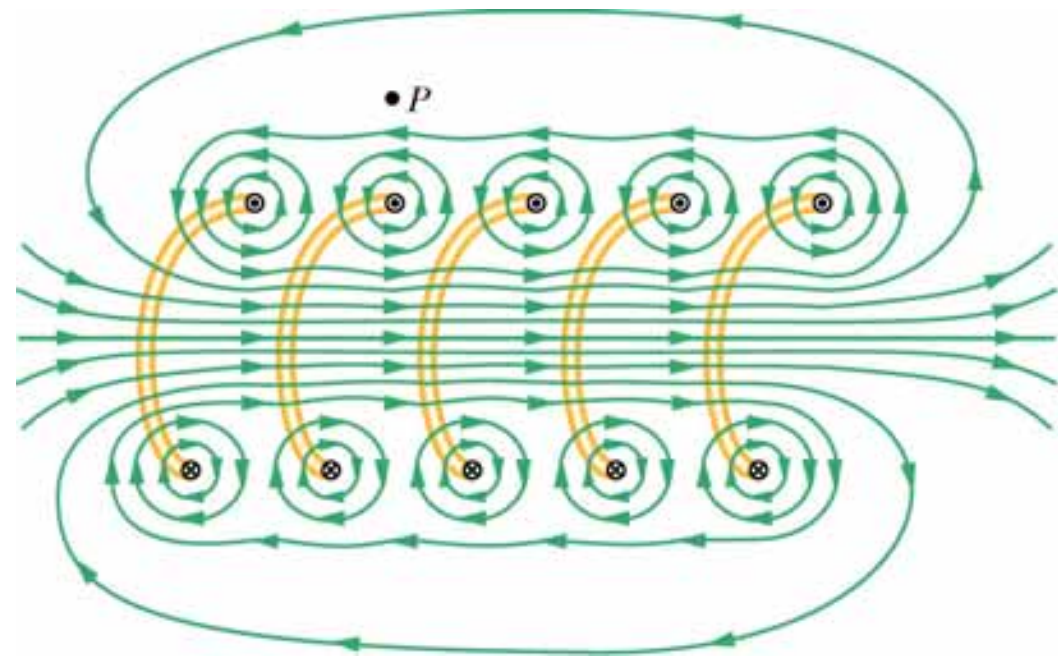
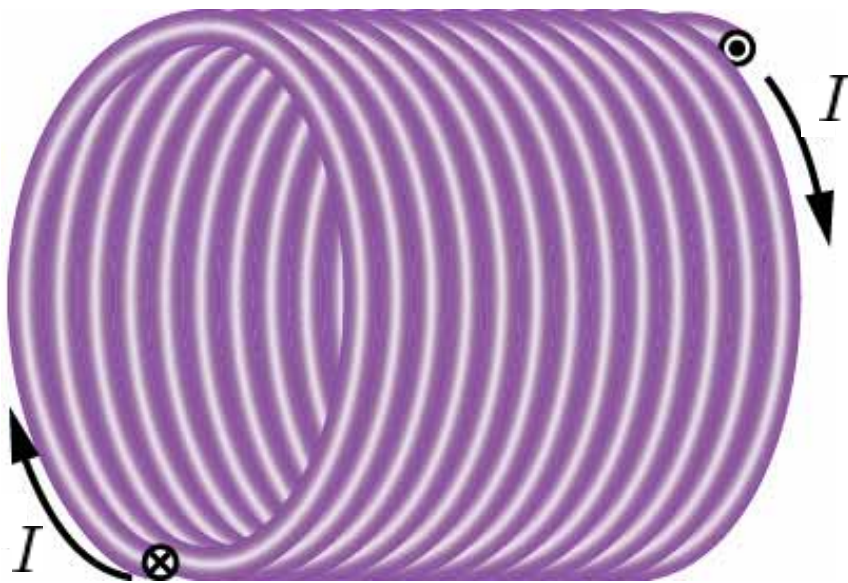


Halliday-Resnick-Walker (2003) Abb. 30.9

## zu 4.4: Kraft zwischen stromdurchflossenen Leitern

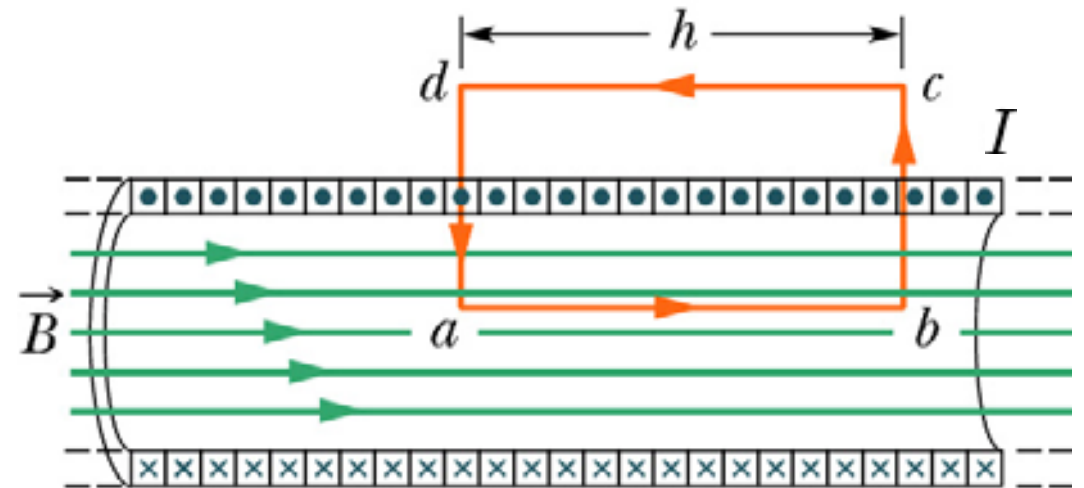
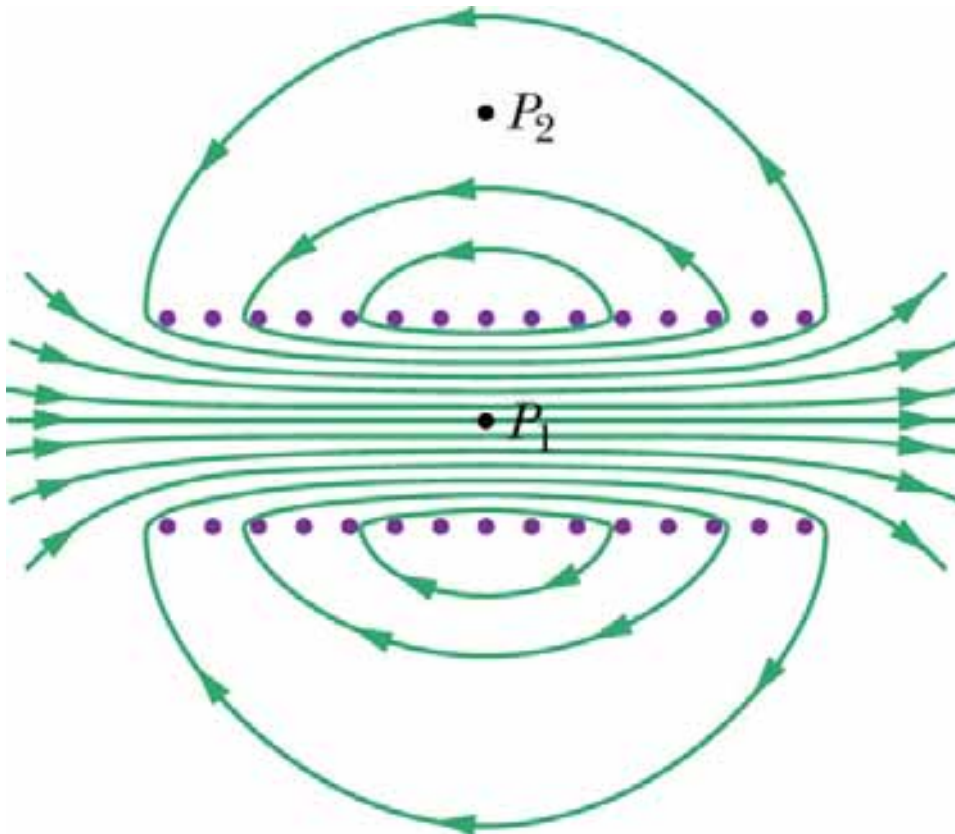


## zu 4.4: Lange Spule (Zylinderspule)



Halliday-Resnick-Walker (2003) Abb. 30.16 und 17

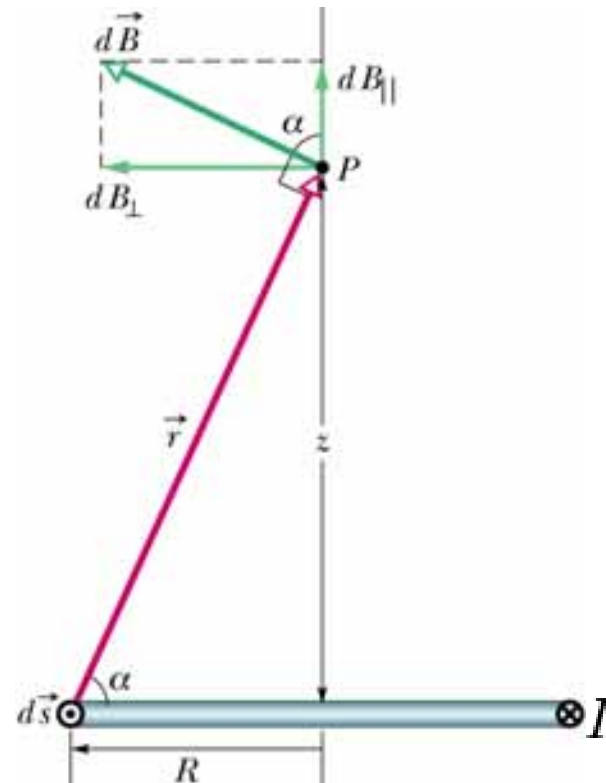
## zu 4.4: Lange Spule (Zylinderspule)



Halliday-Resnick-Walker (2003) Abb. 30.18 und 19

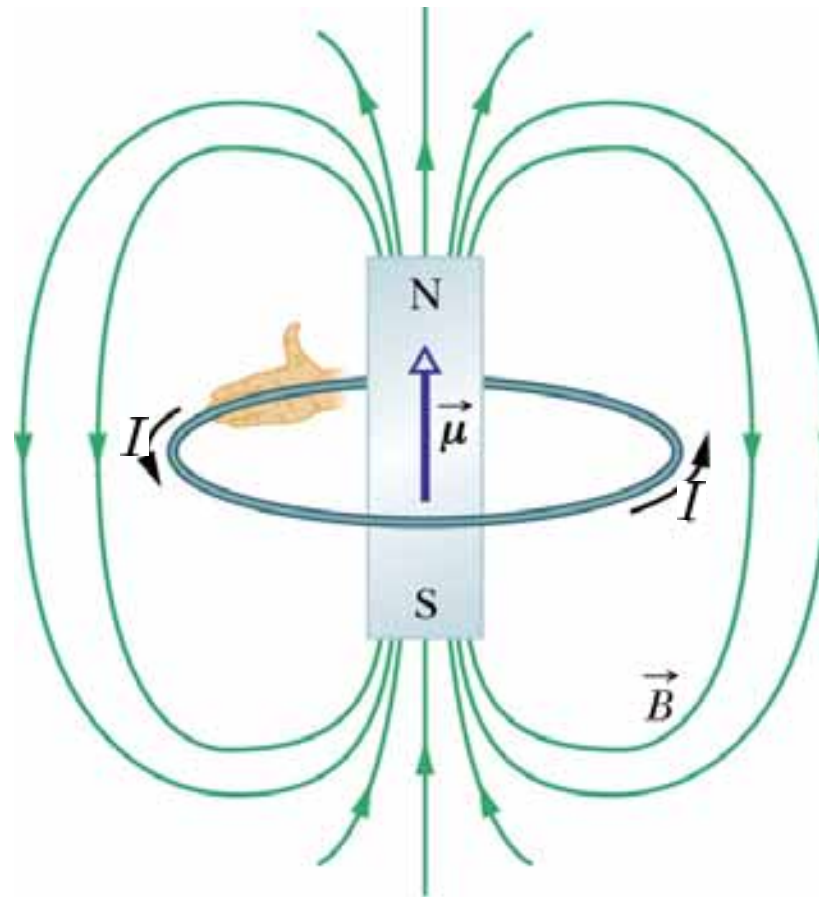


## zu 4.4: Kurze Spule (Ringspule)



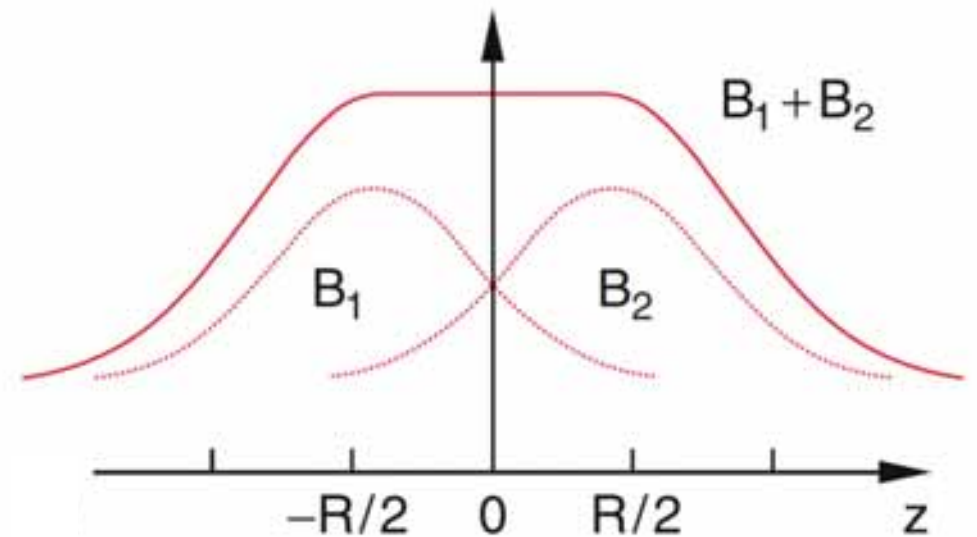
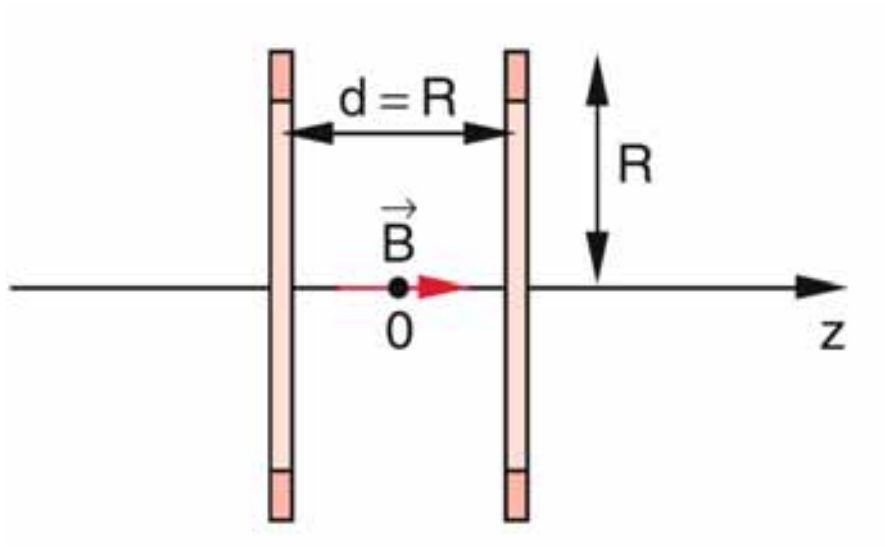
Halliday-Resnick-Walker (2003) Abb. 30.22

## zu 4.4: Kurze Spule (Ringspule)



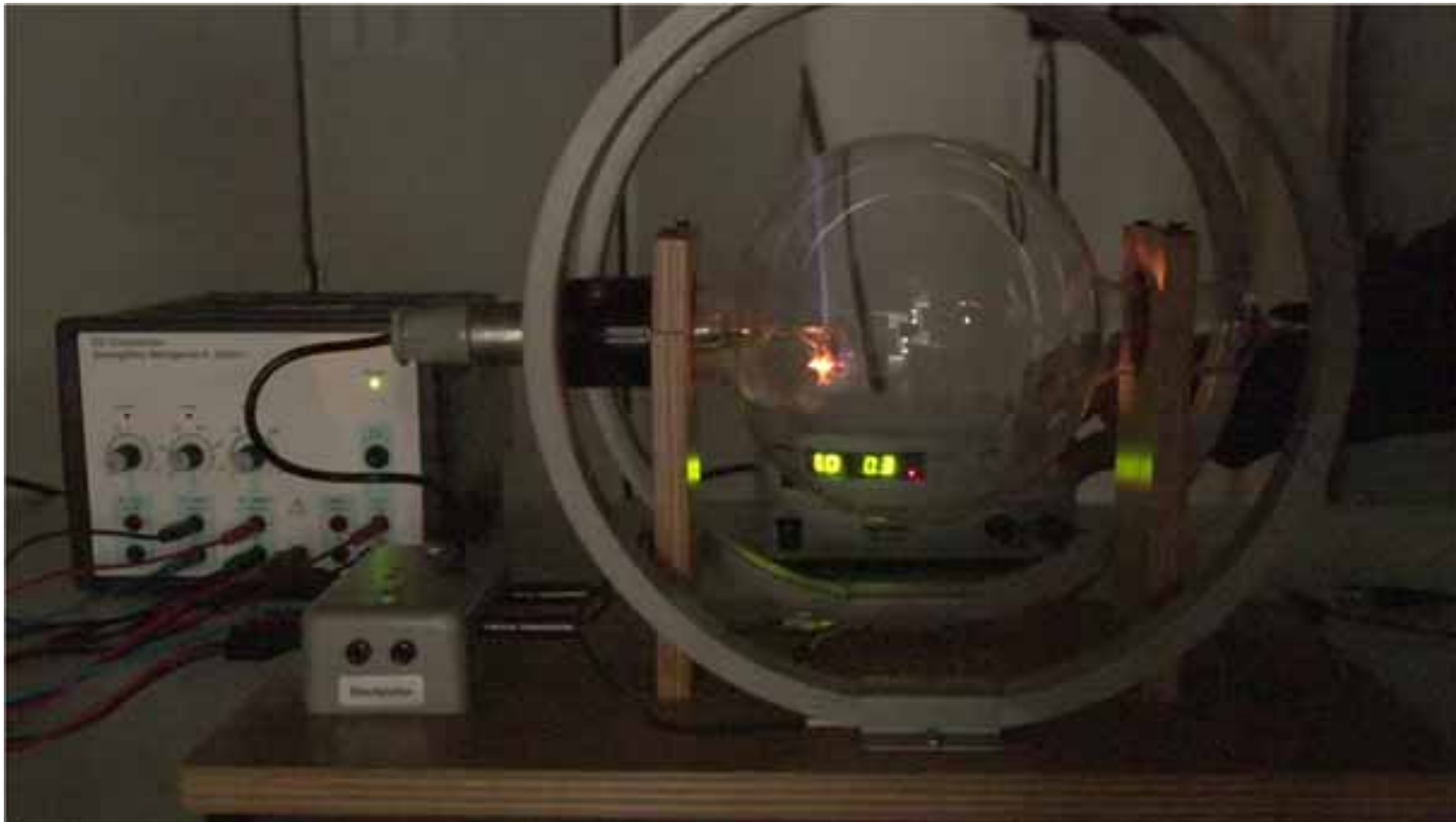
Halliday-Resnick-Walker (2003) Abb. 30.21

## zu 4.4: Helmholtz-Spulenpaar

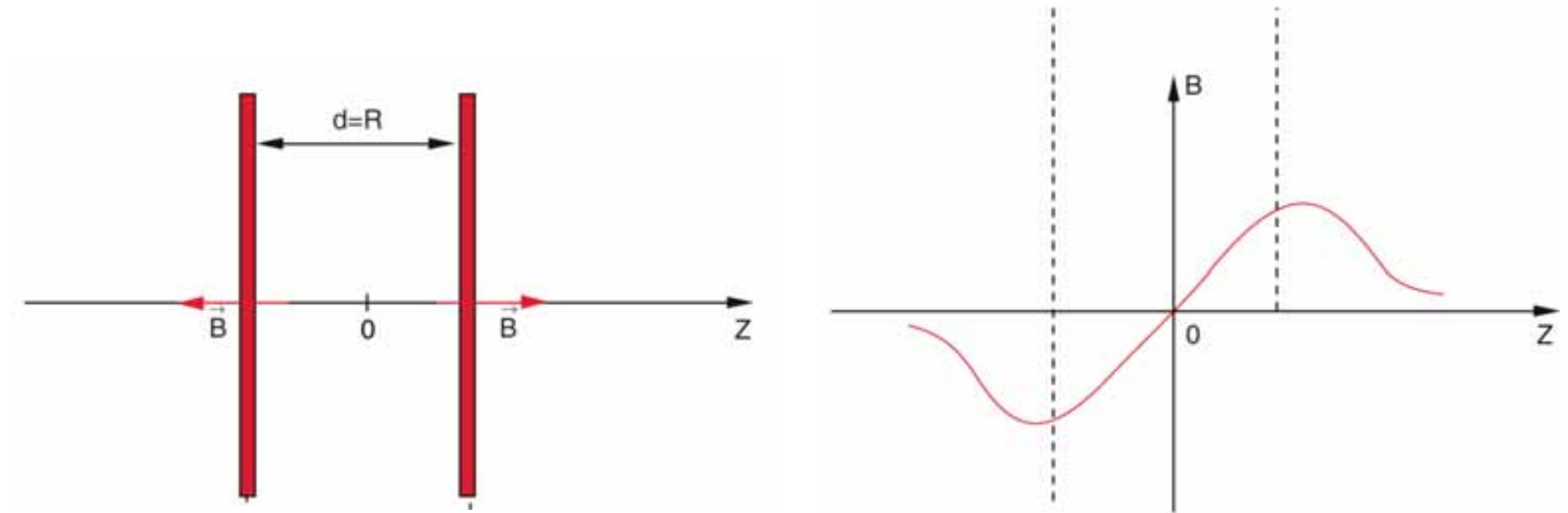


Demtröder – Experimentalphysik 2 (2013), Abb. 3.16

## zu 4.4: Helmholtz-Spulenpaar

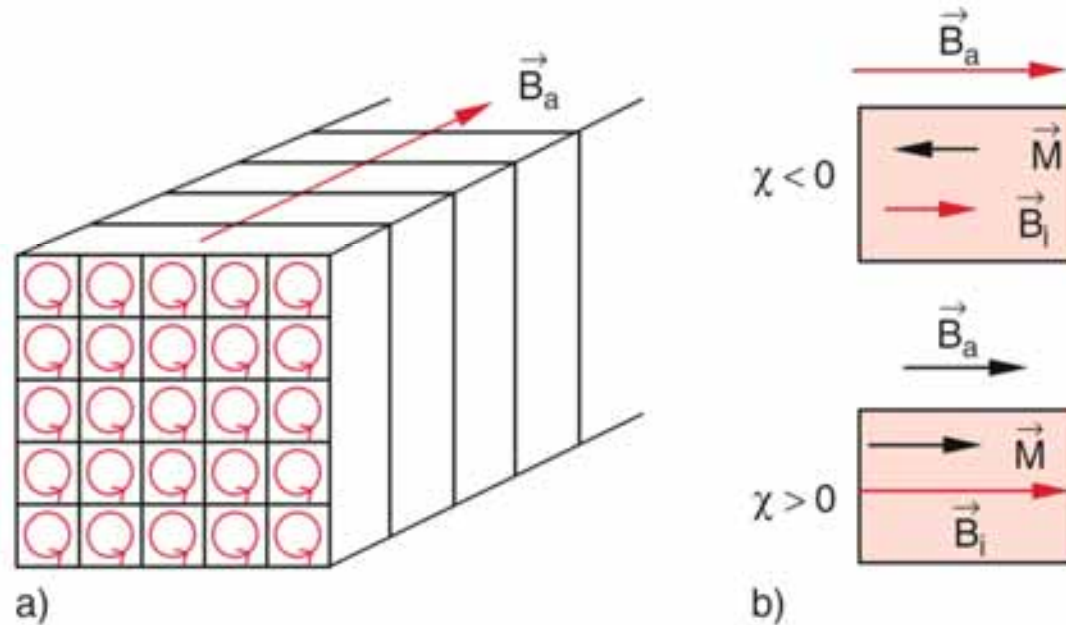


## zu 4.4: Anti-Helmholtz-Spulenpaar



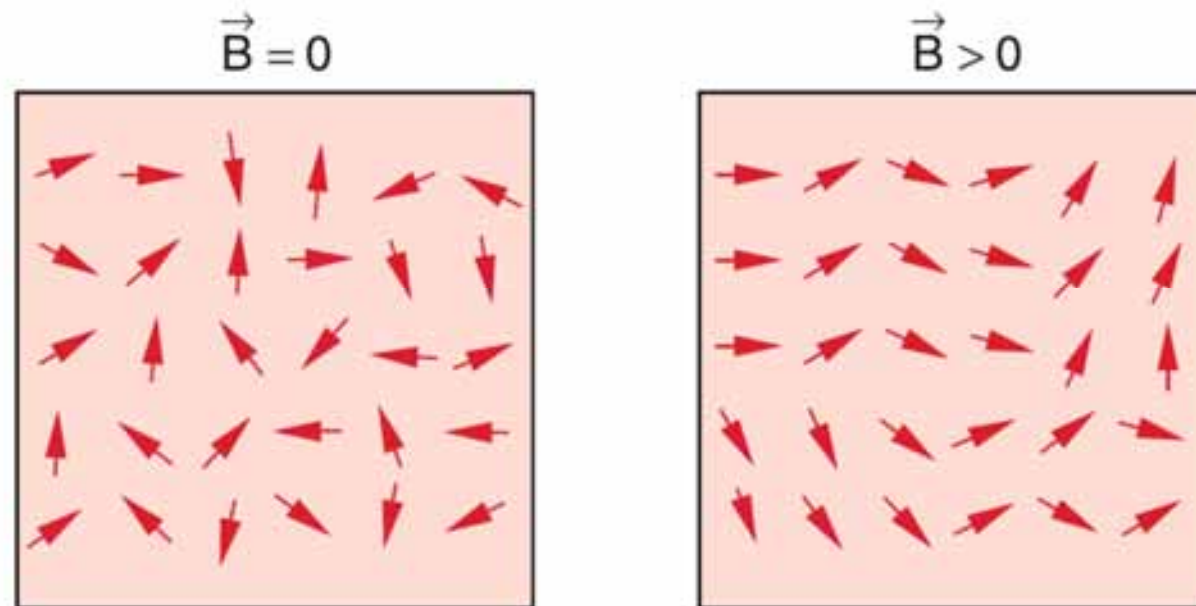
Demtröder – Experimentalphysik 2 (2013), Abb. 3.17

## zu 4.4: Materie im Magnetfeld



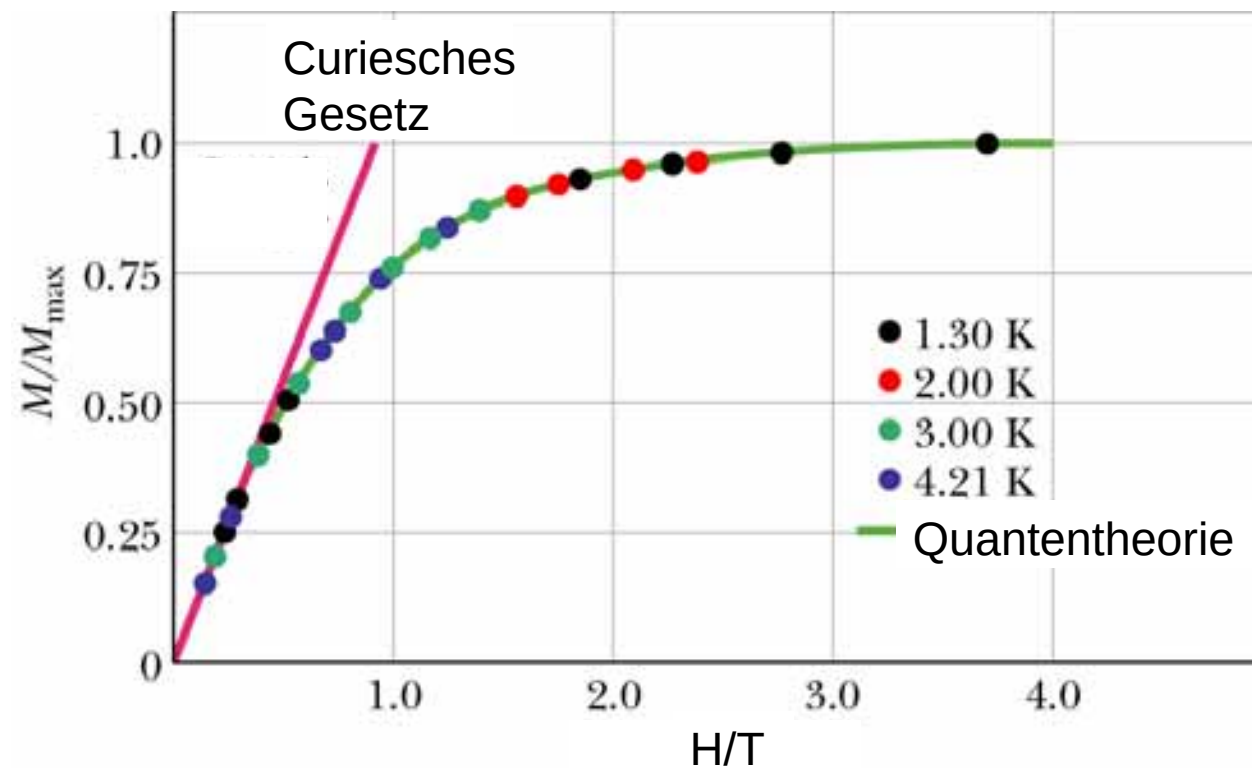
Demtröder – Experimentalphysik 2 (2013), Abb. 3.40

## zu 4.4: Paramagnetismus



Demtröder – Experimentalphysik 2 (2013), Abb. 3.44

## zu 4.4: Paramagnetismus

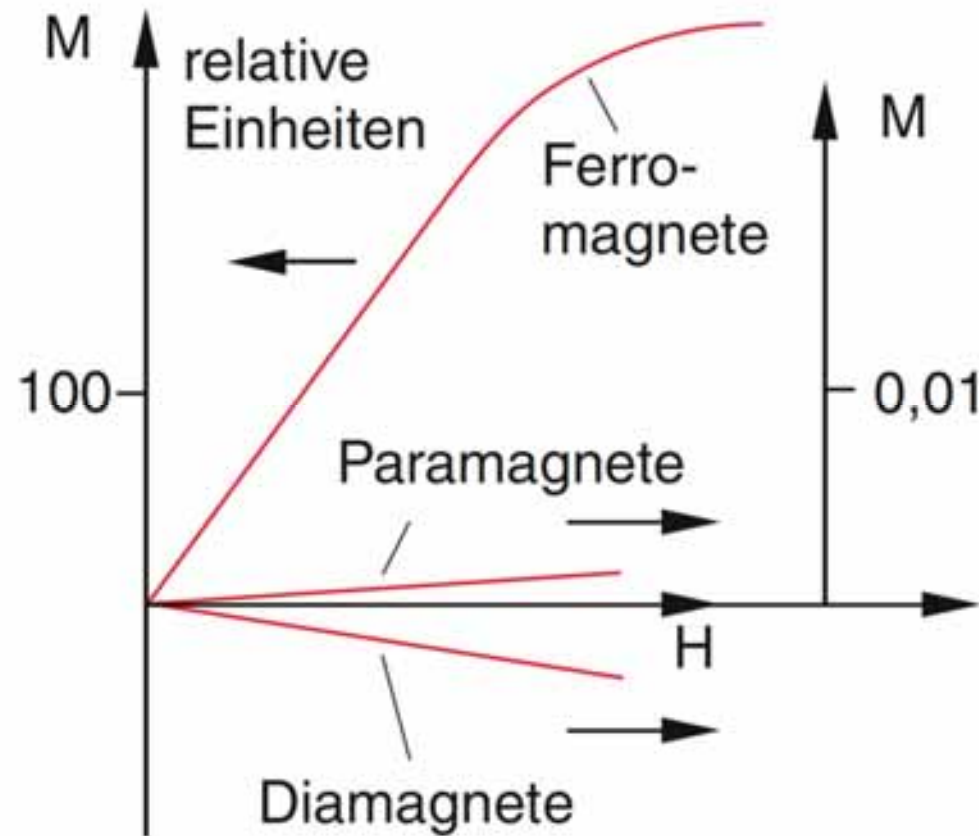


Magnetisierungskurve von Kaliumchromsulfat

Halliday-Resnick-Walker (2003) Abb. 32.9

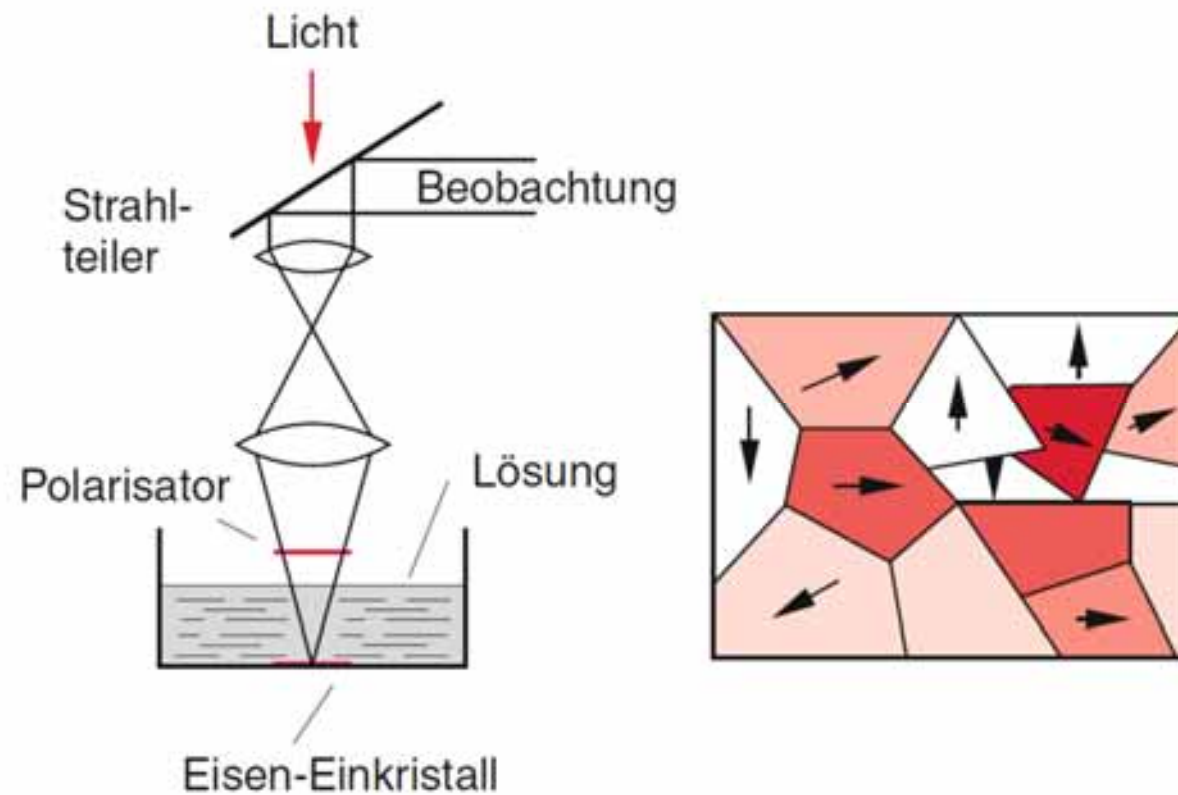


## zu 4.4: Materie im Magnetfeld



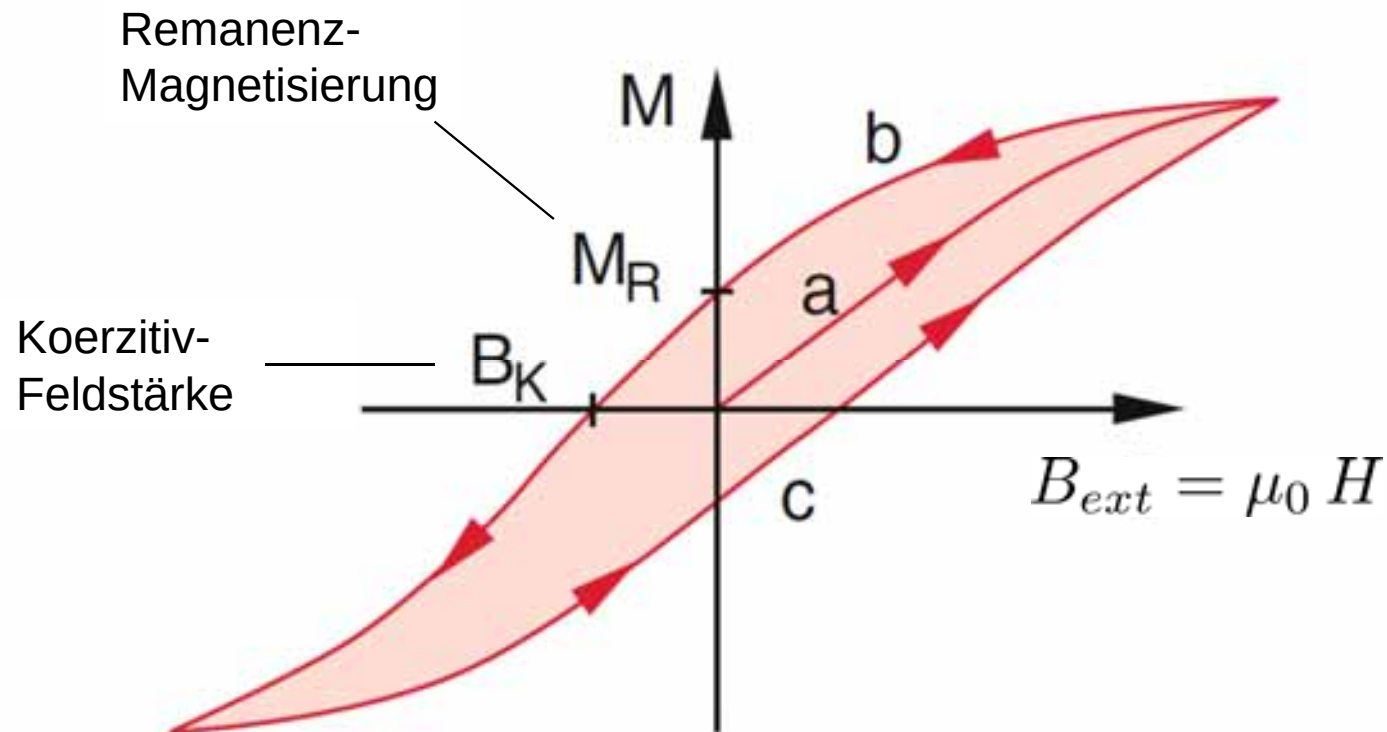
Demtröder – Experimentalphysik 2 (2013), Abb. 3.41

## zu 4.4: Ferromagnetismus – Weißsche Bezirke



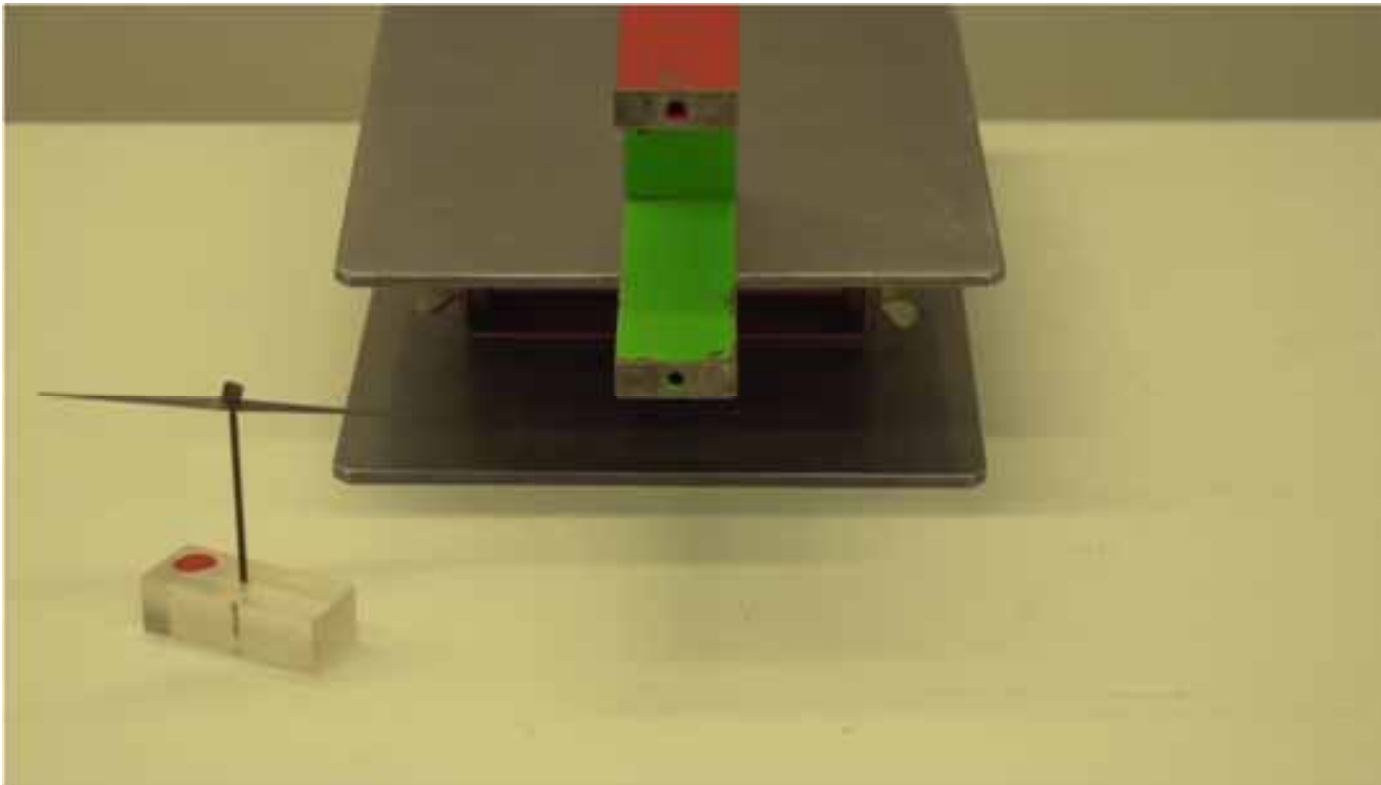
Demtröder – Experimentalphysik 2 (2013), Abb. 3.49

## zu 4.4: Ferromagnetismus - Hysterese



Demtröder – Experimentalphysik 2 (2013), Abb. 3.41

## zu 4.4: Permanentmagnete



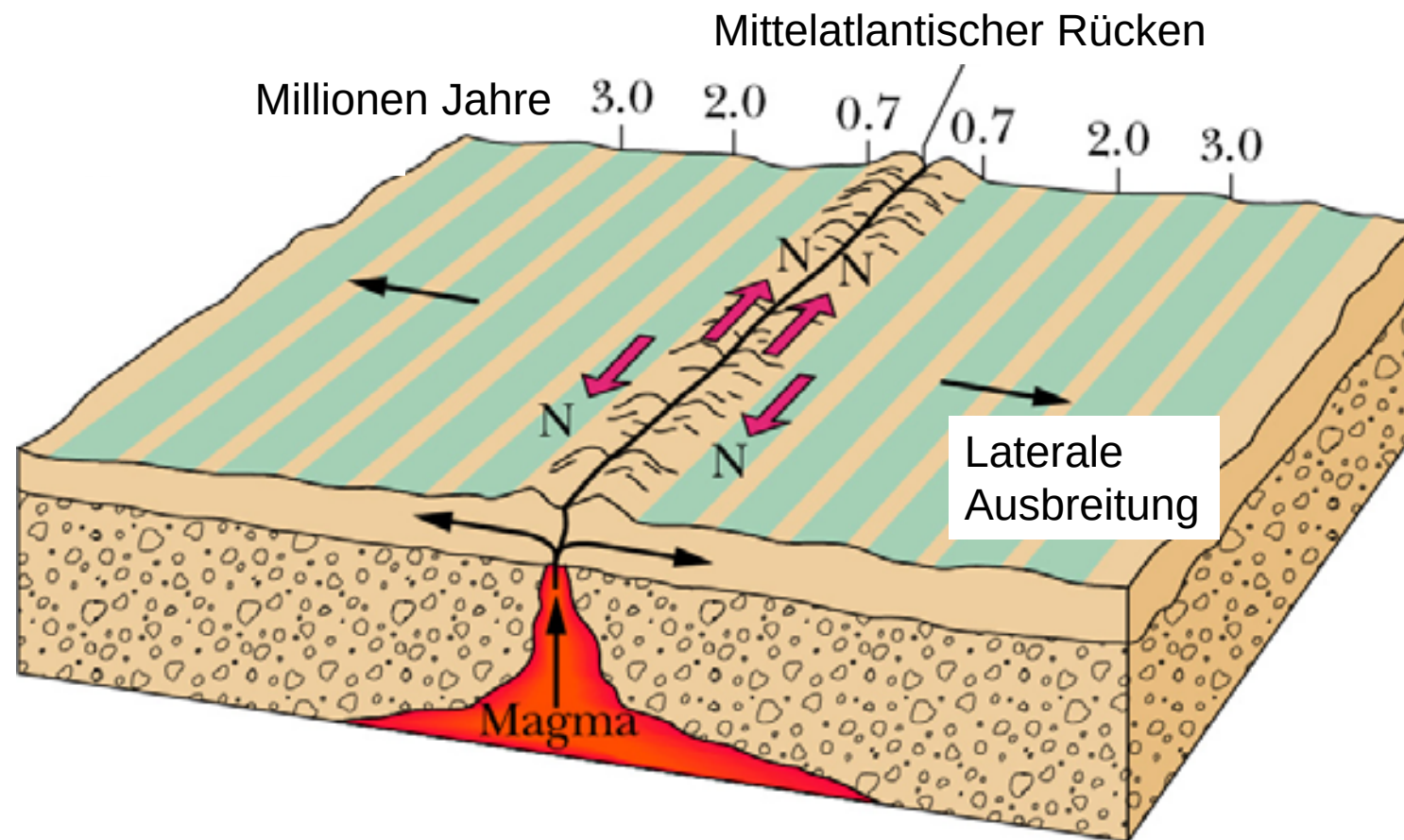
Demtröder – Experimentalphysik 2 (2013), Abb. 3.49

## zu 4.4: Permanentmagnete



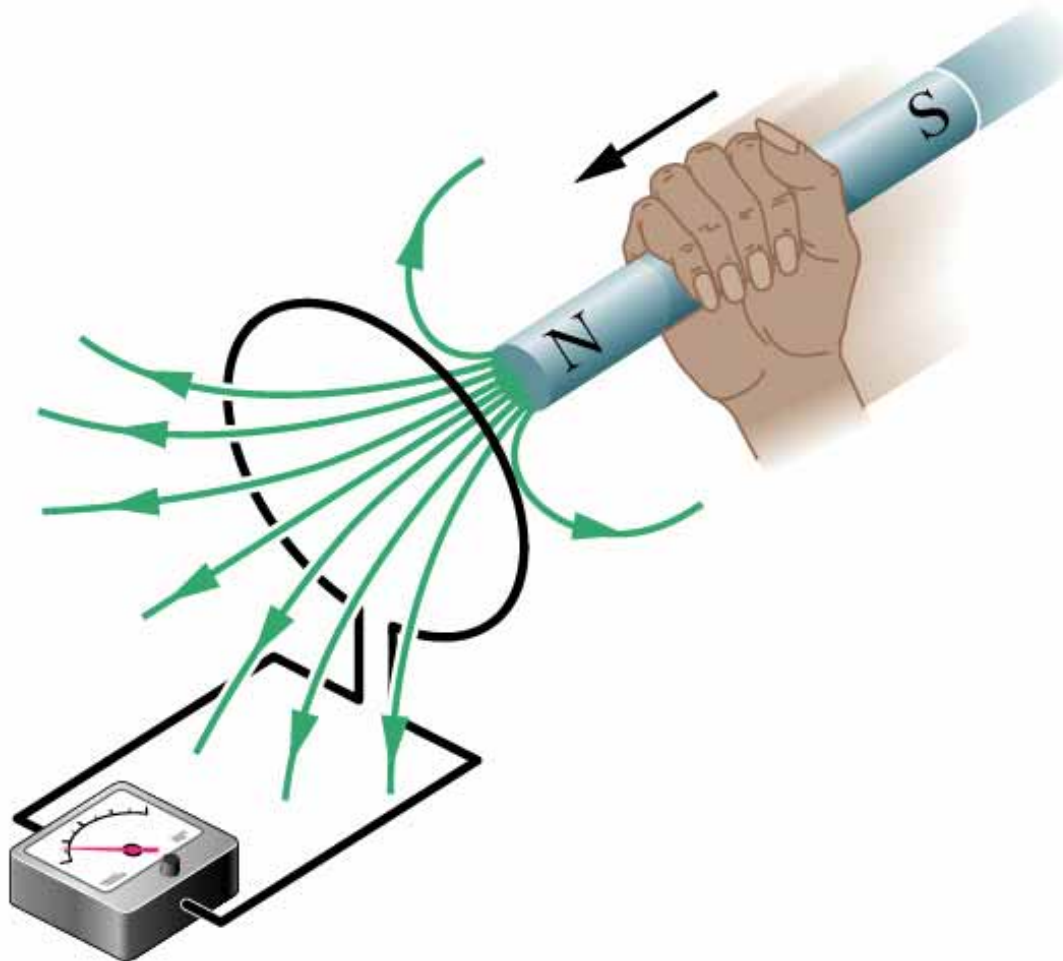
[www.wikipedia.de](http://www.wikipedia.de)

## zu 4.4: Magnetisches Gedächtnis



Halliday-Resnick-Walker (2003) Abb. 32-5

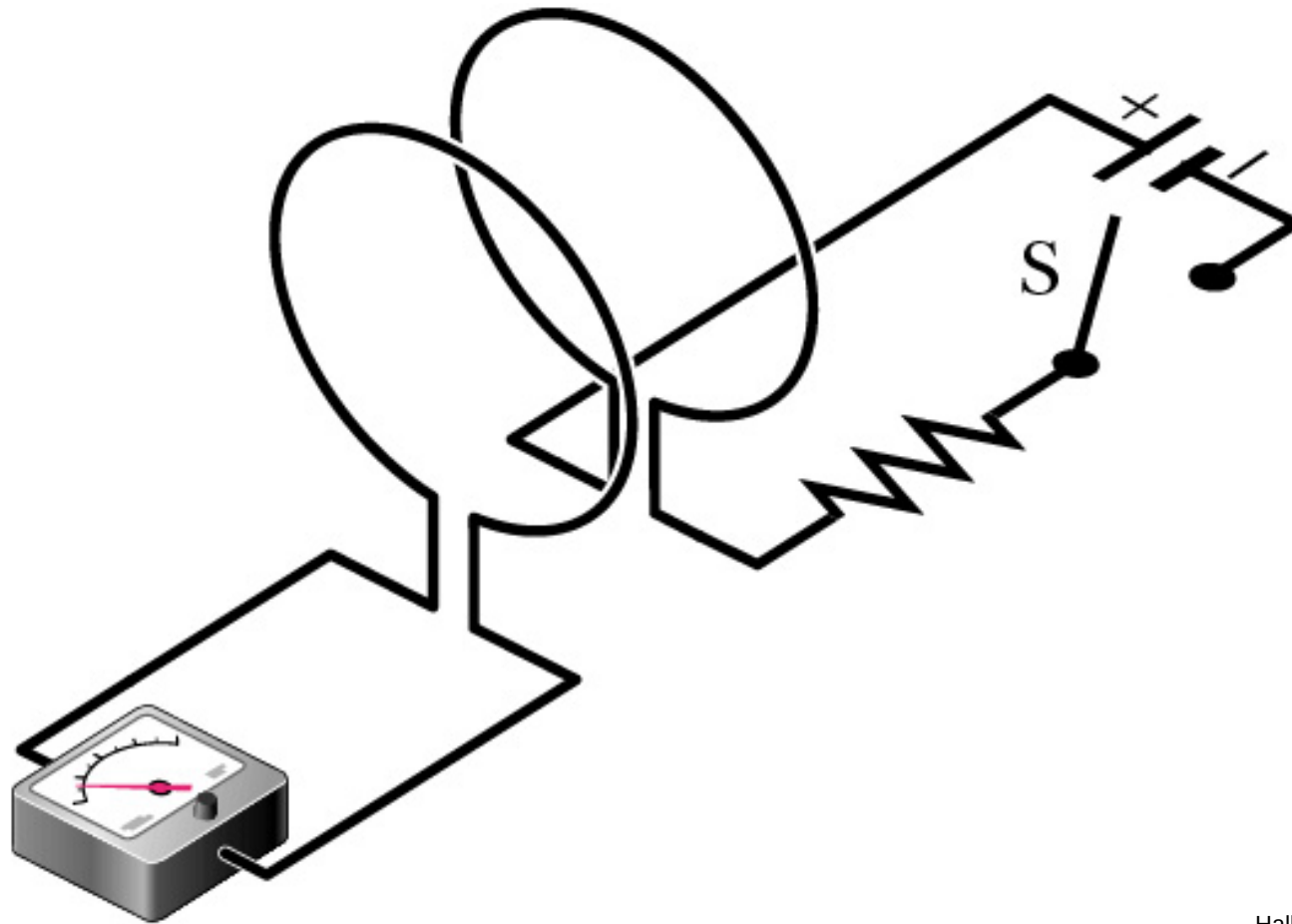
## zu 4.5: Magnetische Induktion



Halliday-Resnick-Walker (2003) Abb. 31-1



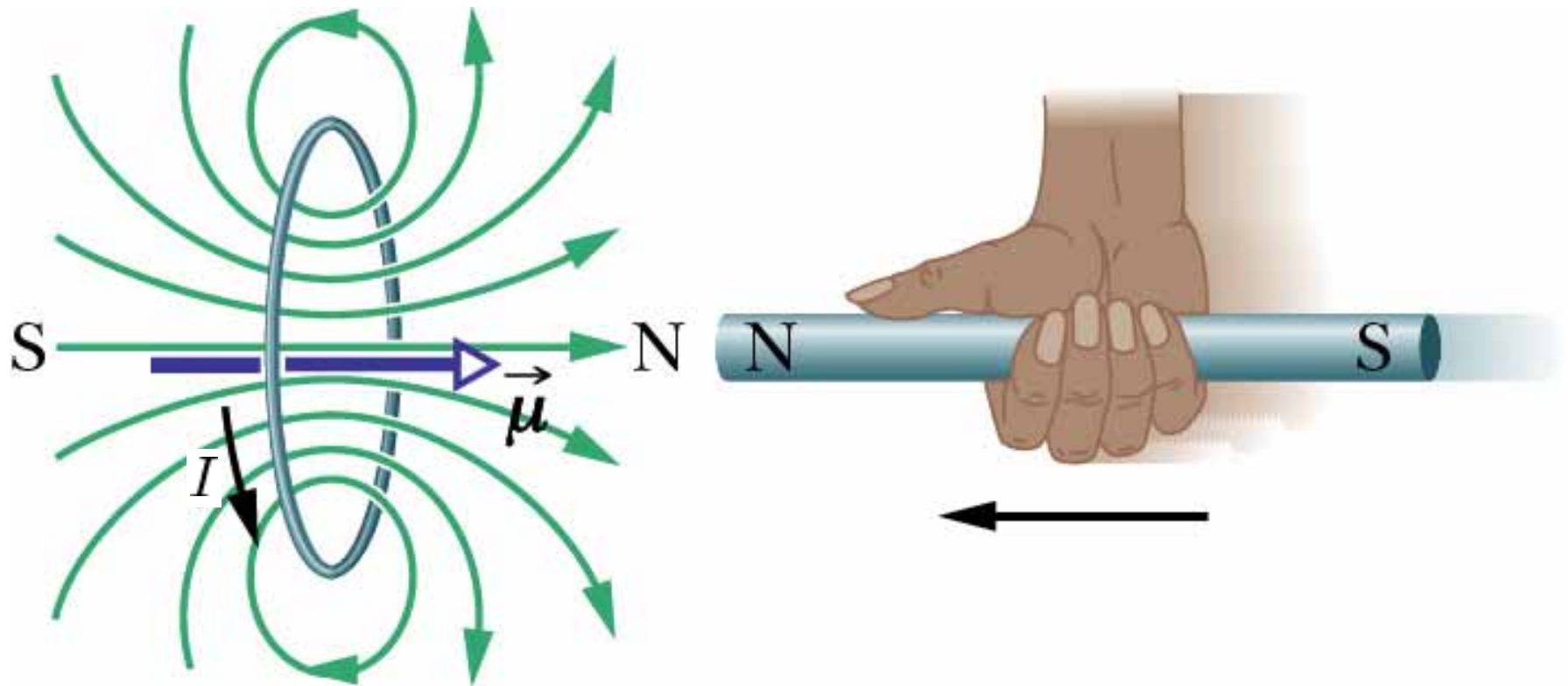
## zu 4.5: Magnetische Induktion



Halliday-Resnick-Walker (2003) Abb. 31-2

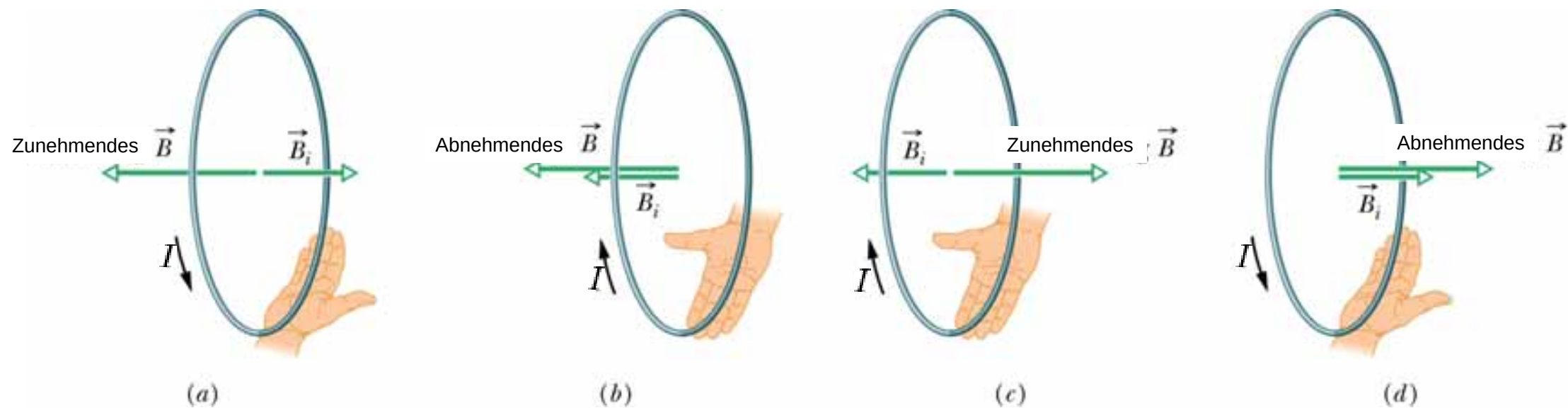


## zu 4.5: Lenzsche Regel



Halliday-Resnick-Walker (2003) Abb. 31-4

# zu 4.5: Lenzsche Regel

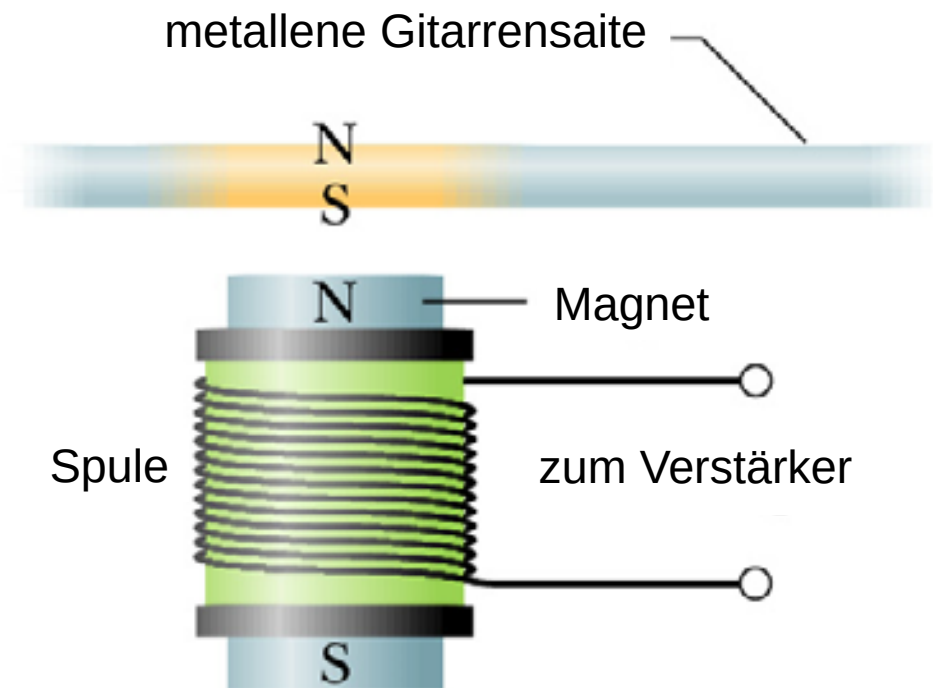


Halliday-Resnick-Walker (2003) Abb. 31-5

## zu 4.5: Lenzsche Regel

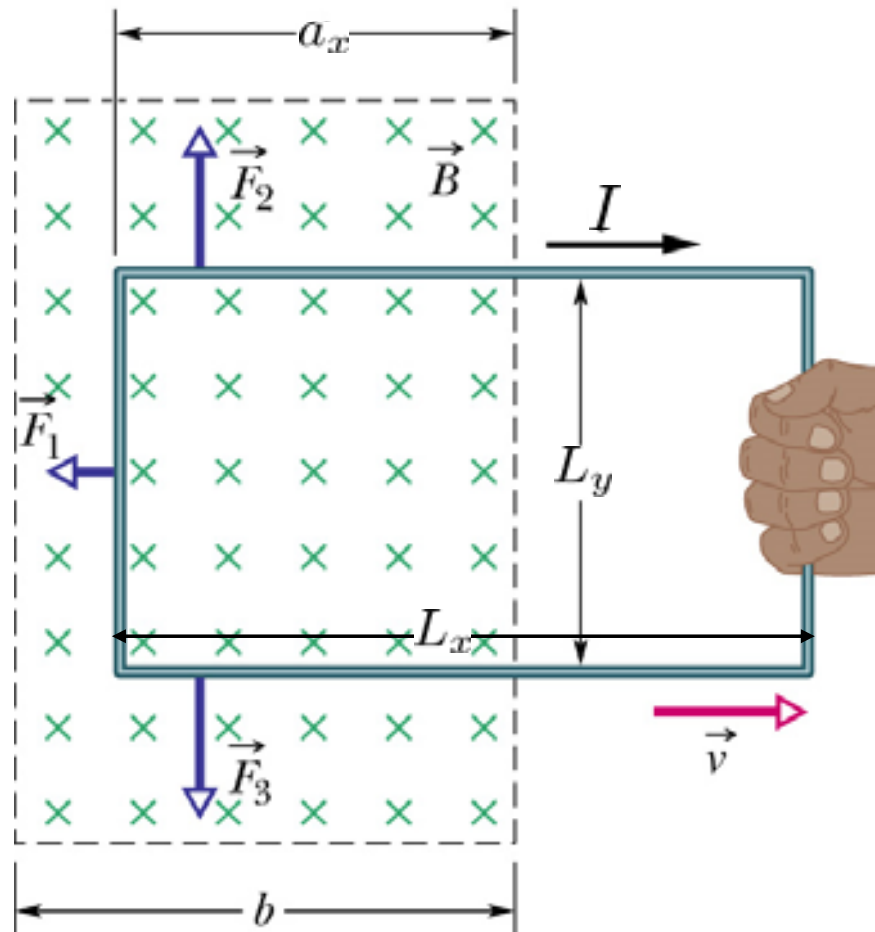


[www.wikipedia.de](http://www.wikipedia.de)



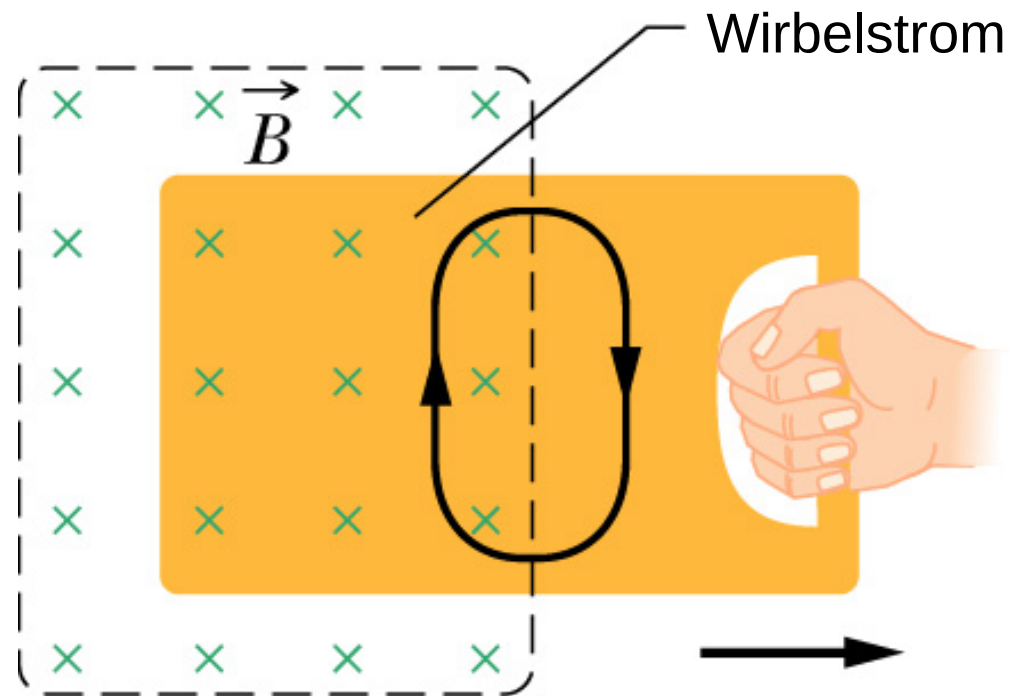
Halliday-Resnick-Walker (2003) Abb. 31-7

## zu 4.5: Induktion und Energietransfer



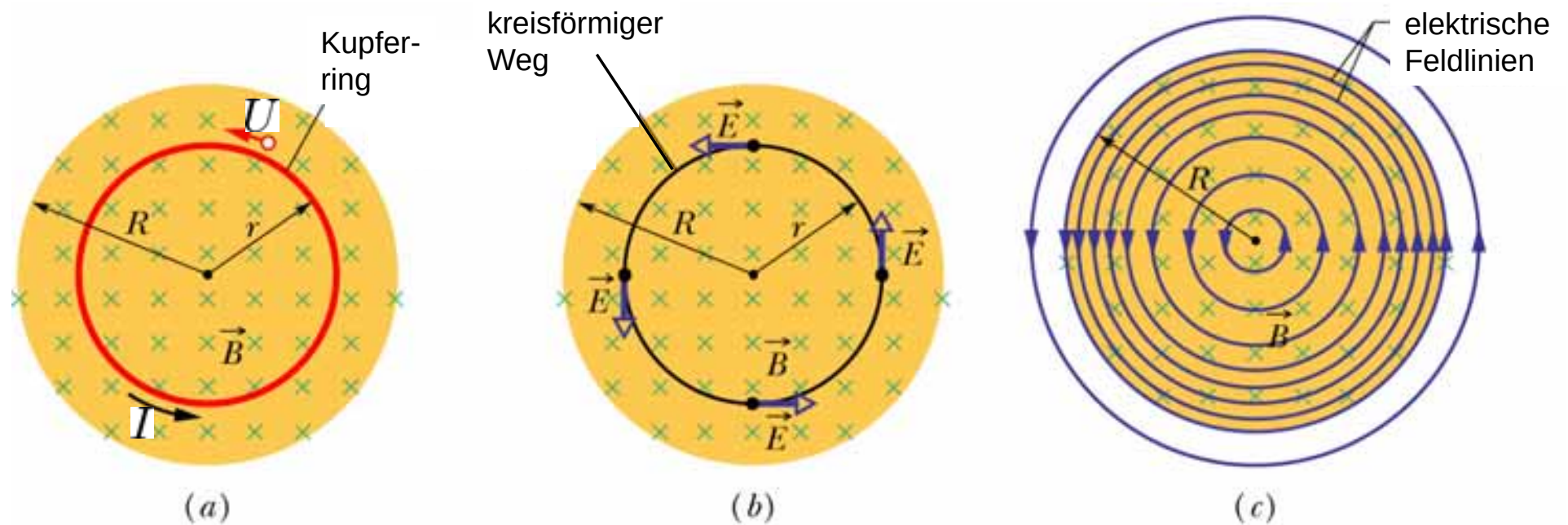
Halliday-Resnick-Walker (2003) Abb. 31.10

## zu 4.5: Wirbelstrom



Halliday-Resnick-Walker (2003) Abb. 31.12

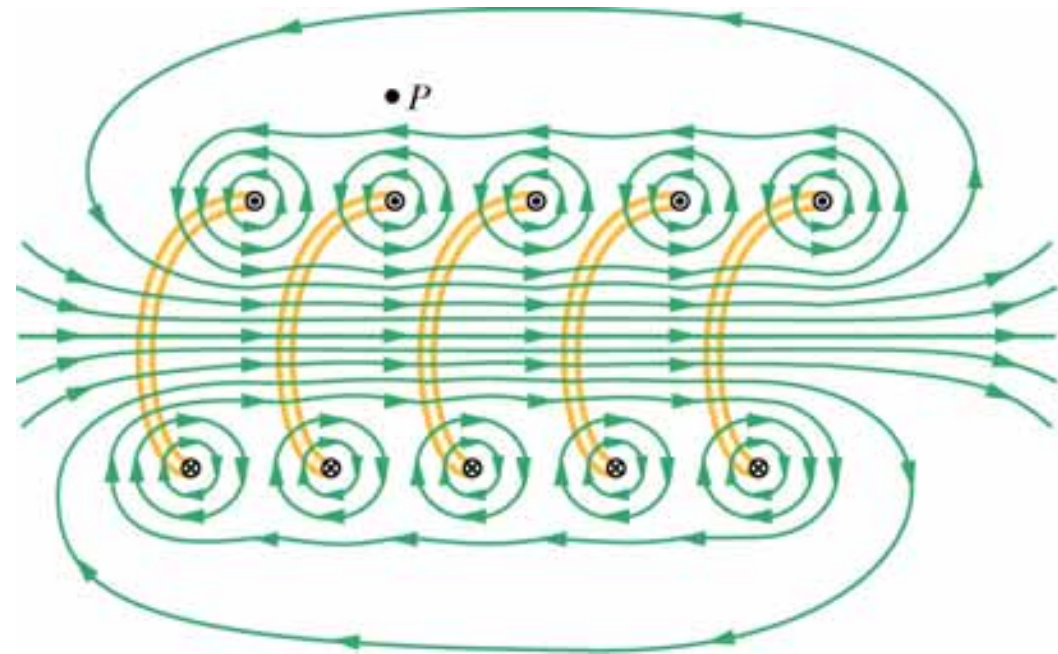
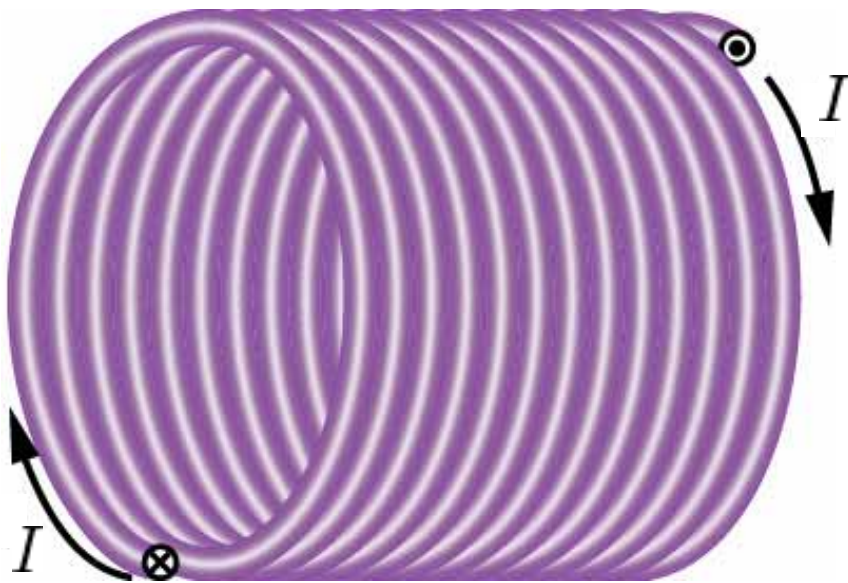
## zu 4.5: Induzierte elektrische Felder



Halliday-Resnick-Walker (2003) Abb. 31.13

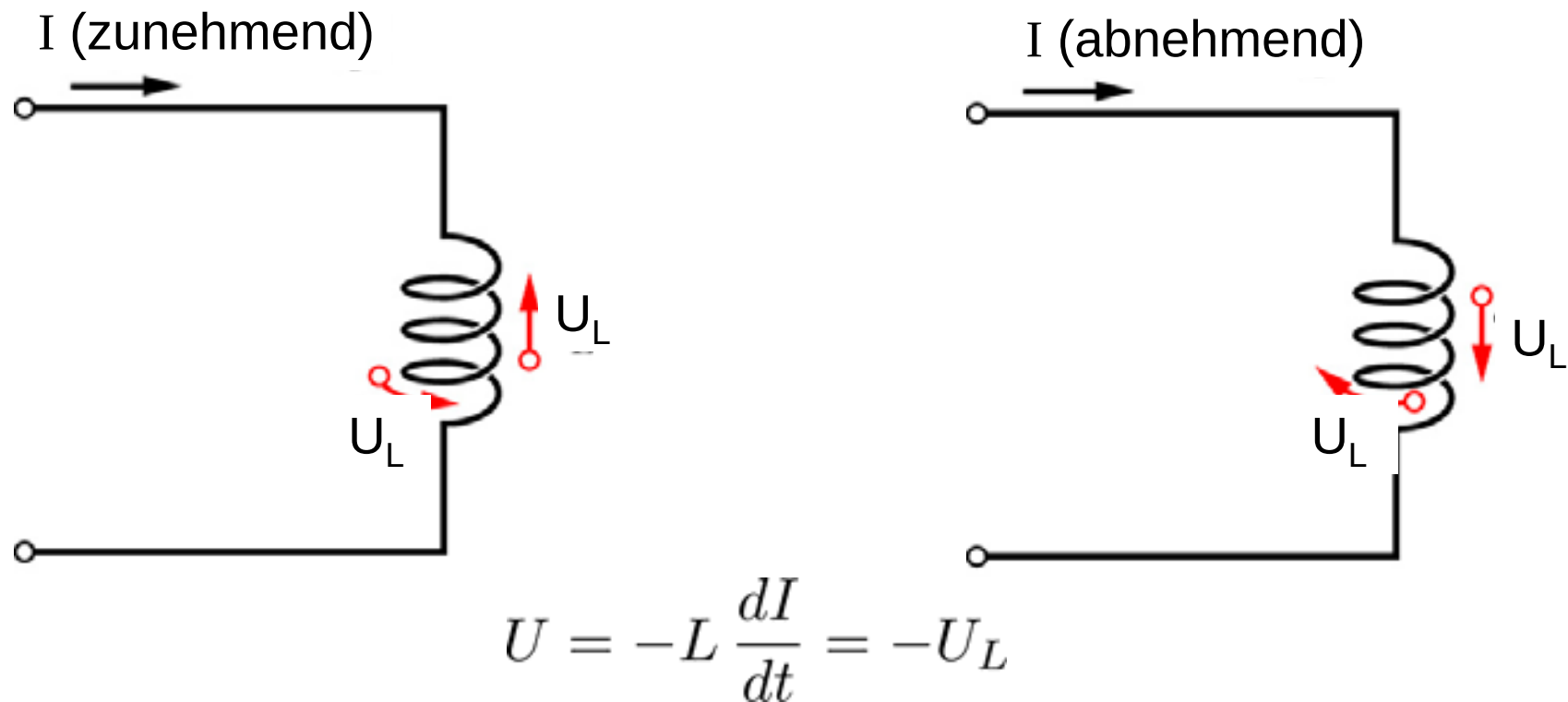


## zu 4.4: Lange Spule (Zylinderspule)



Halliday-Resnick-Walker (2003) Abb. 30.16 und 17

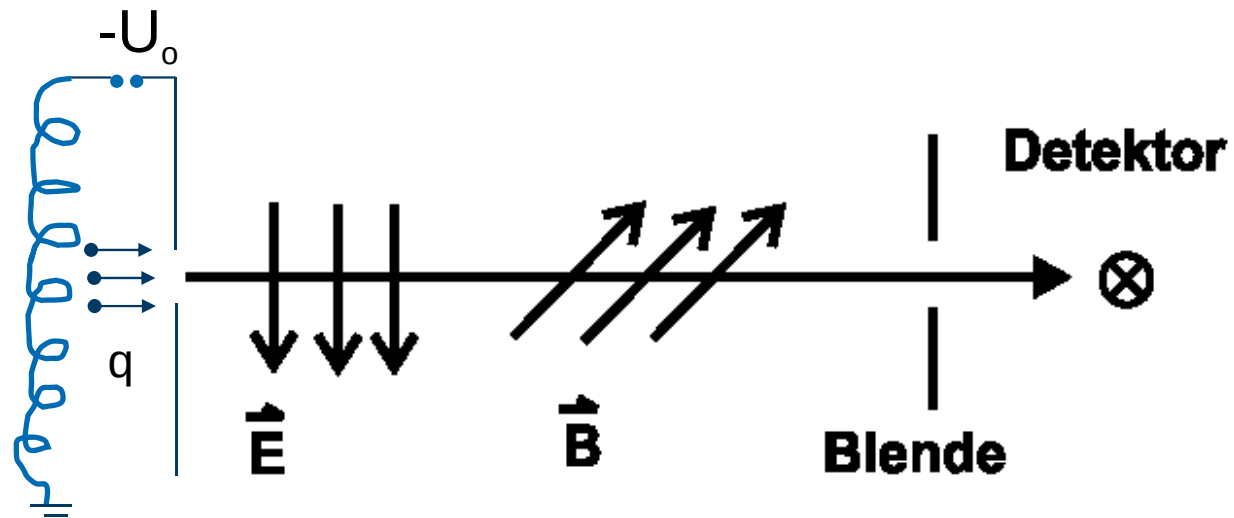
## zu 2.5 (4.5): Induktivität einer Spule



Halliday-Resnick-Walker (2003) Abb. 31.16

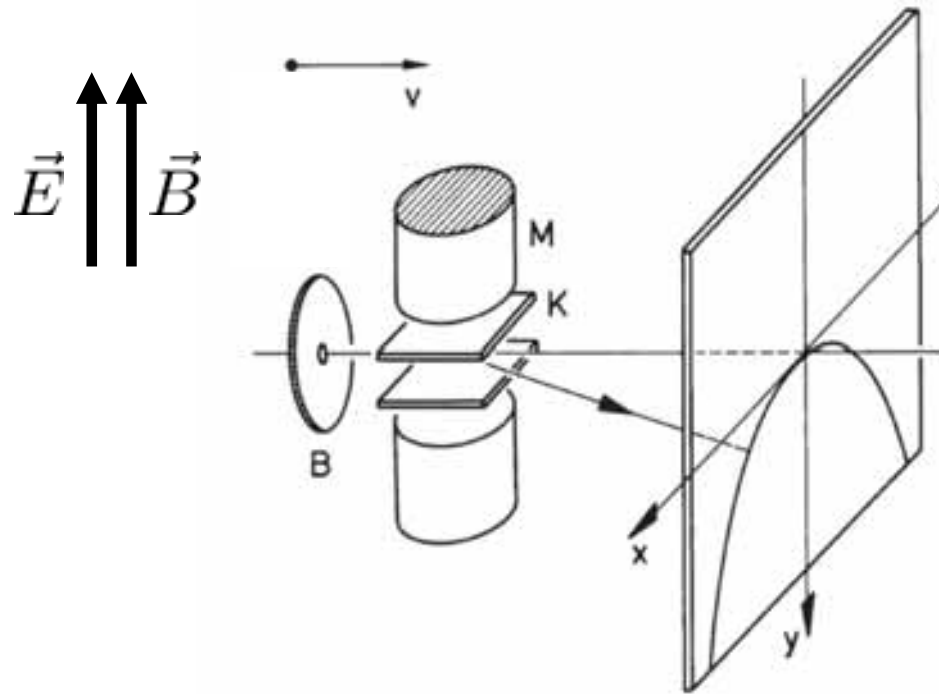


## zu 4.6: Wien-Filter



Skript, Experimentalphysik III, Uni Tübingen, S. 53

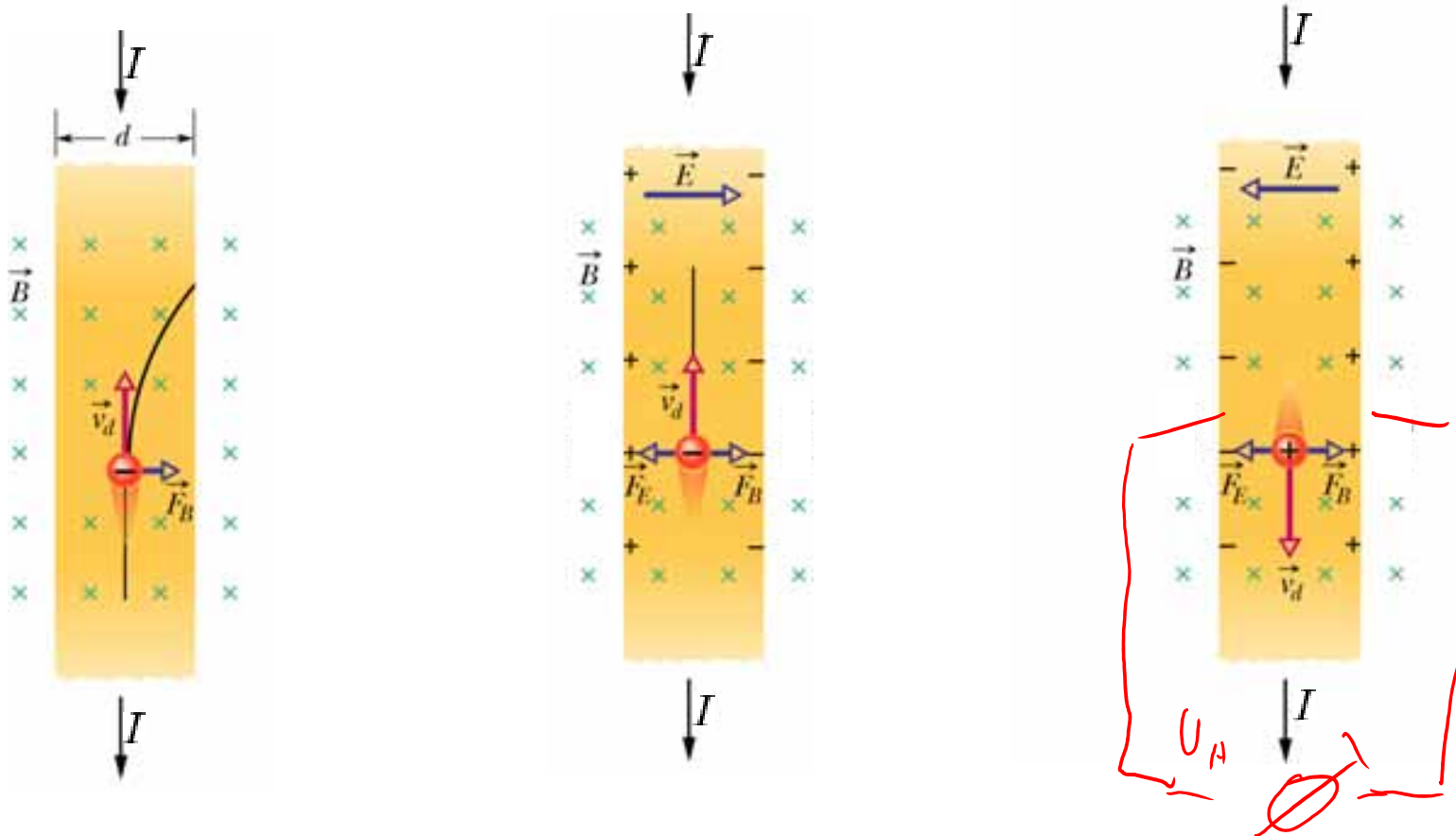
## zu 4.6: Massenspektrometrie



Parabelmethode zur Bestimmung der spezifischen Ladung

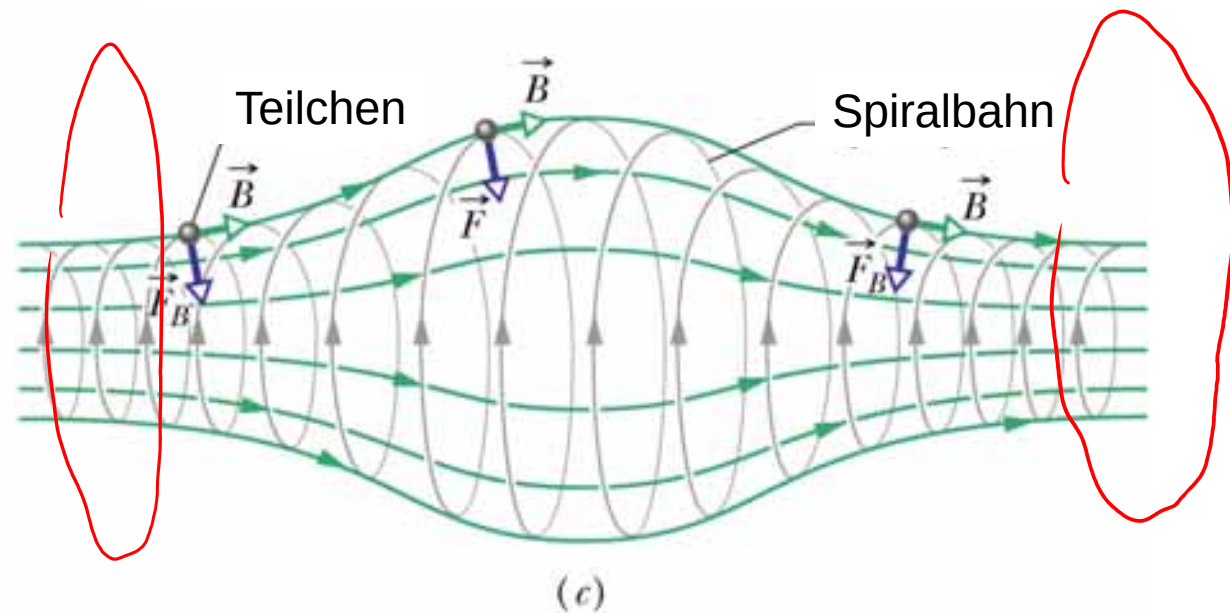
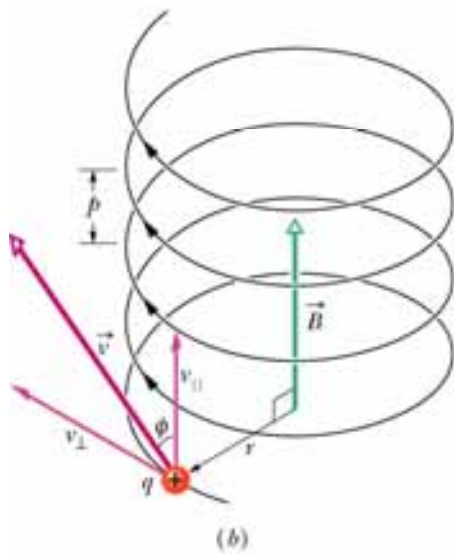
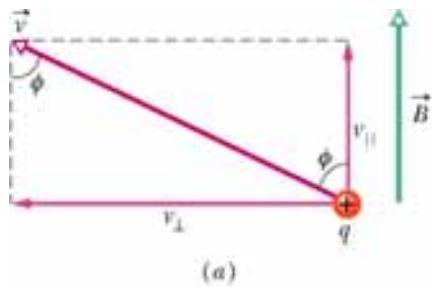
Haken-Wolf, Abb. 3.2

## zu 4.6: Hall-Effekt



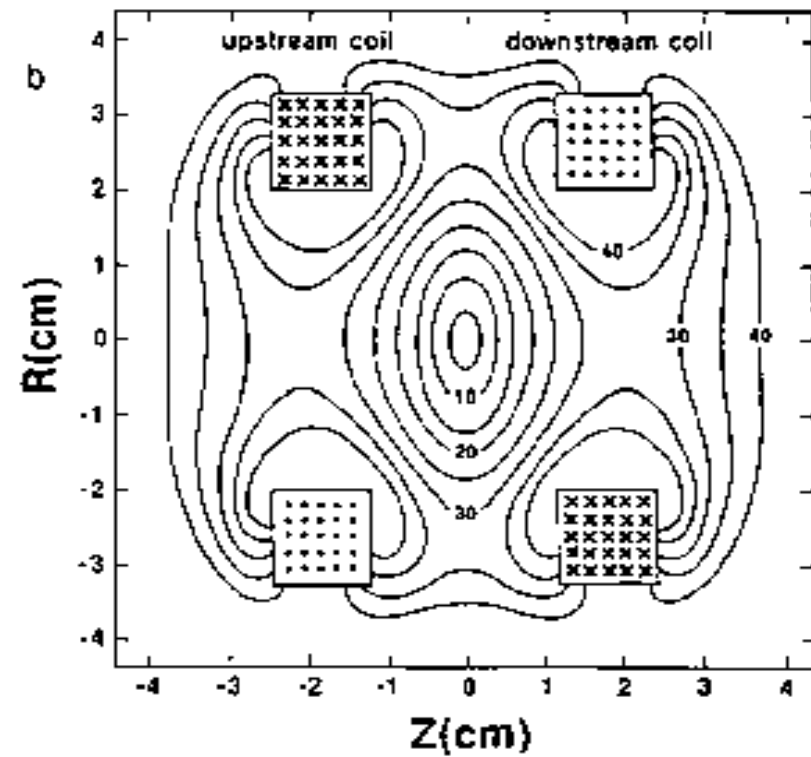
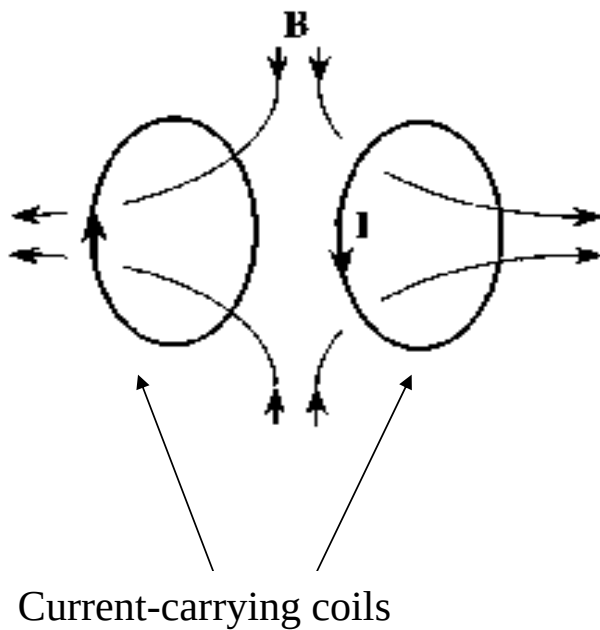
Halliday-Resnick-Walker (2003) Abb. 29.8

## zu 4.6: Magnetische Flasche



Halliday-Resnick-Walker (2003) Abb. 29.11

## zu 4.6: Magnetfalle für Neutralatome

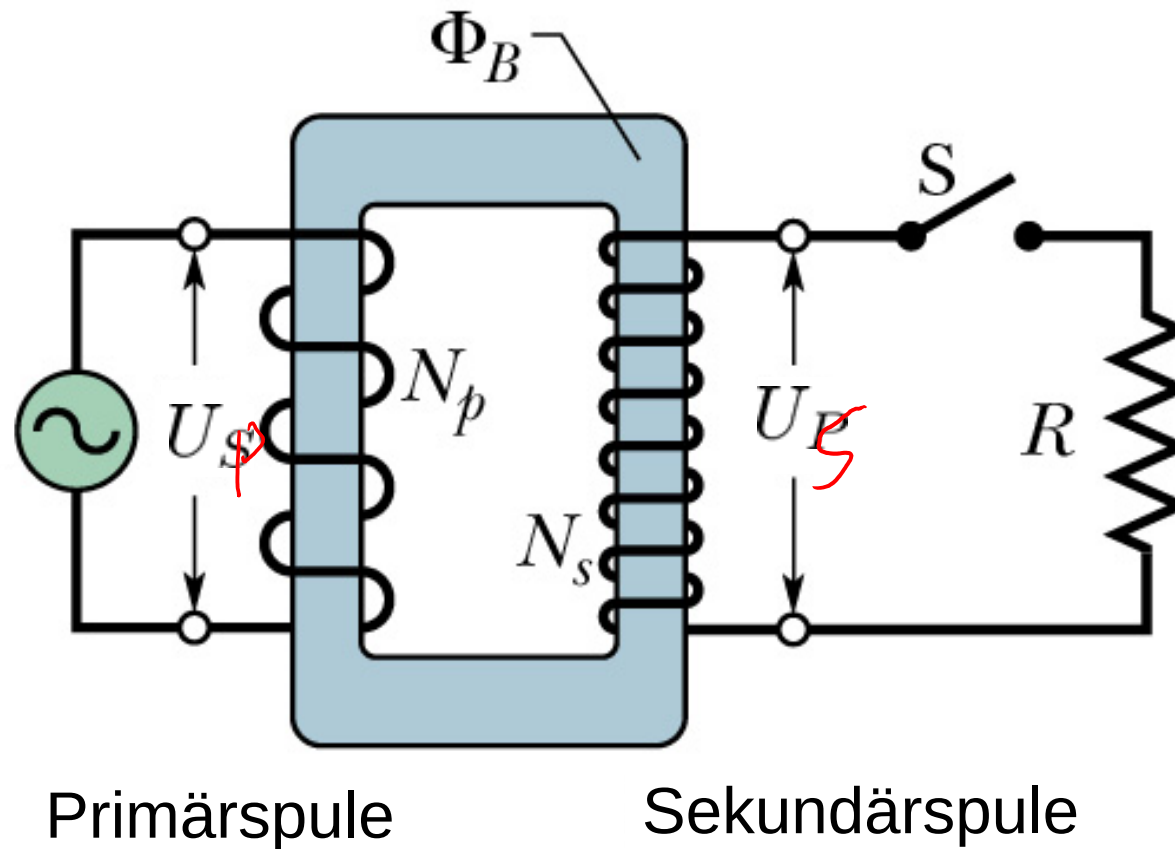


W.D. Phillips, Rev. Mod. Phys. **70**, 721

## zu 4.6: Magnetfalle für Neutralatome



## zu 4.6: Transformator



Halliday-Resnick-Walker (2003) Abb. 33.15



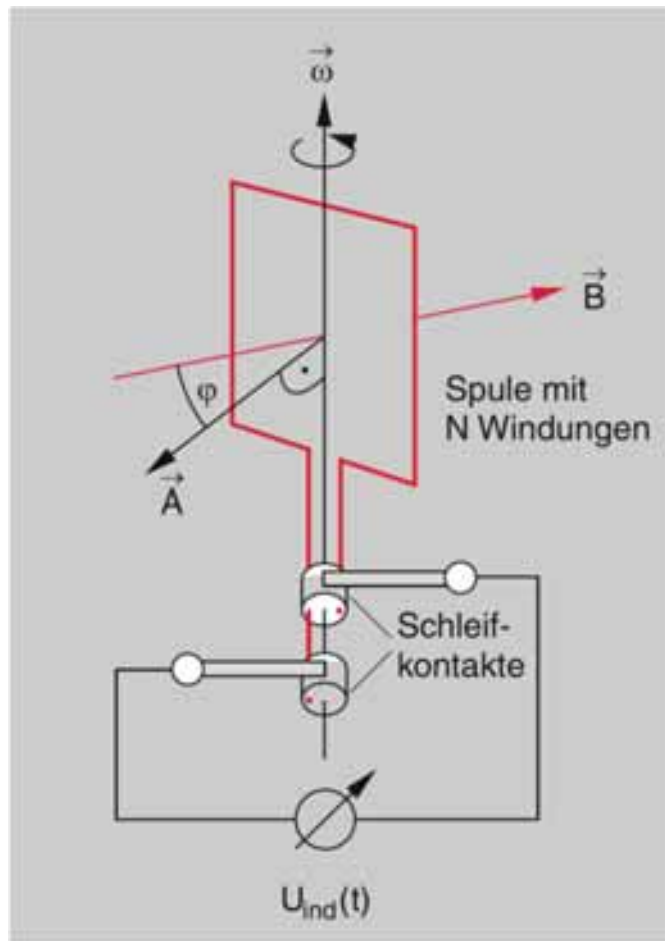
## zu 4.6: Transformator



Demtröder – Experimentalphysik 2 (2013), Abb. 5.39

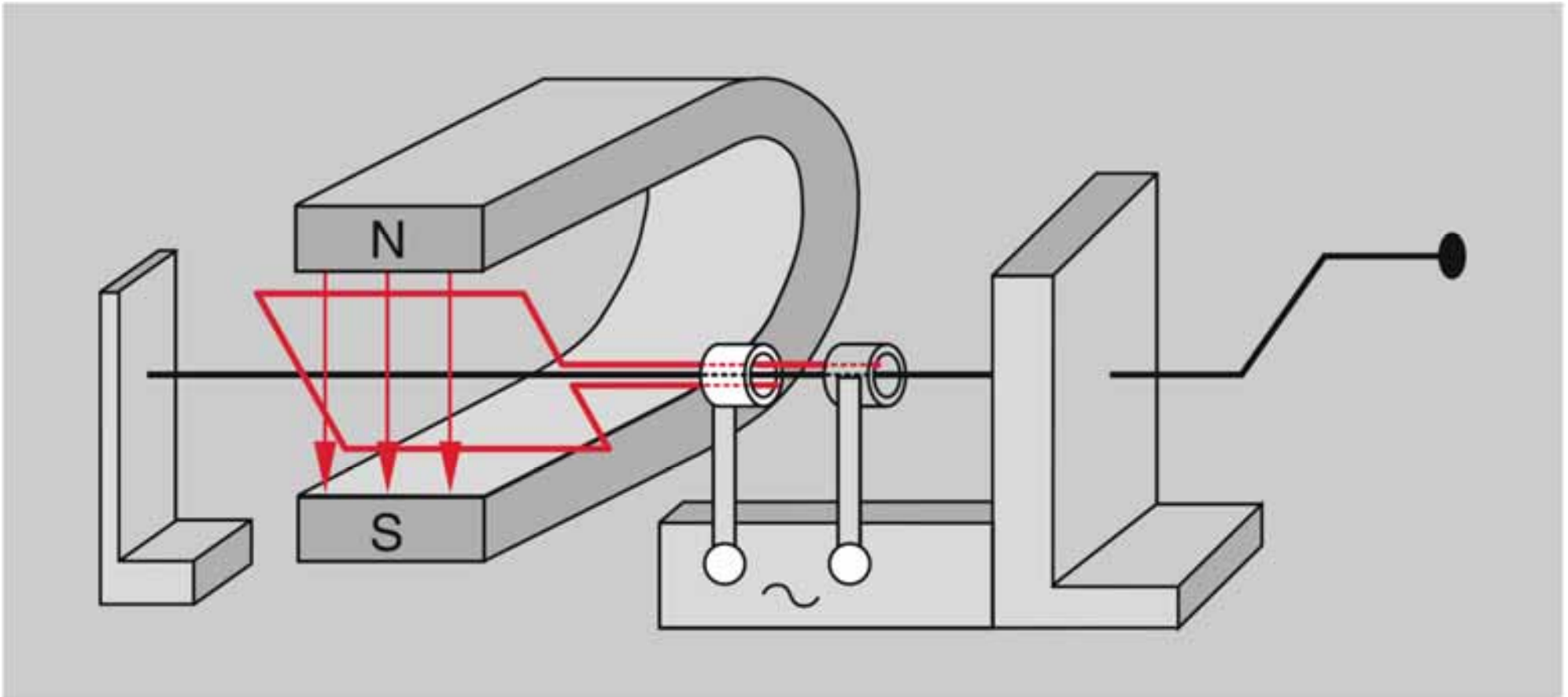


## zu 4.6: Generator



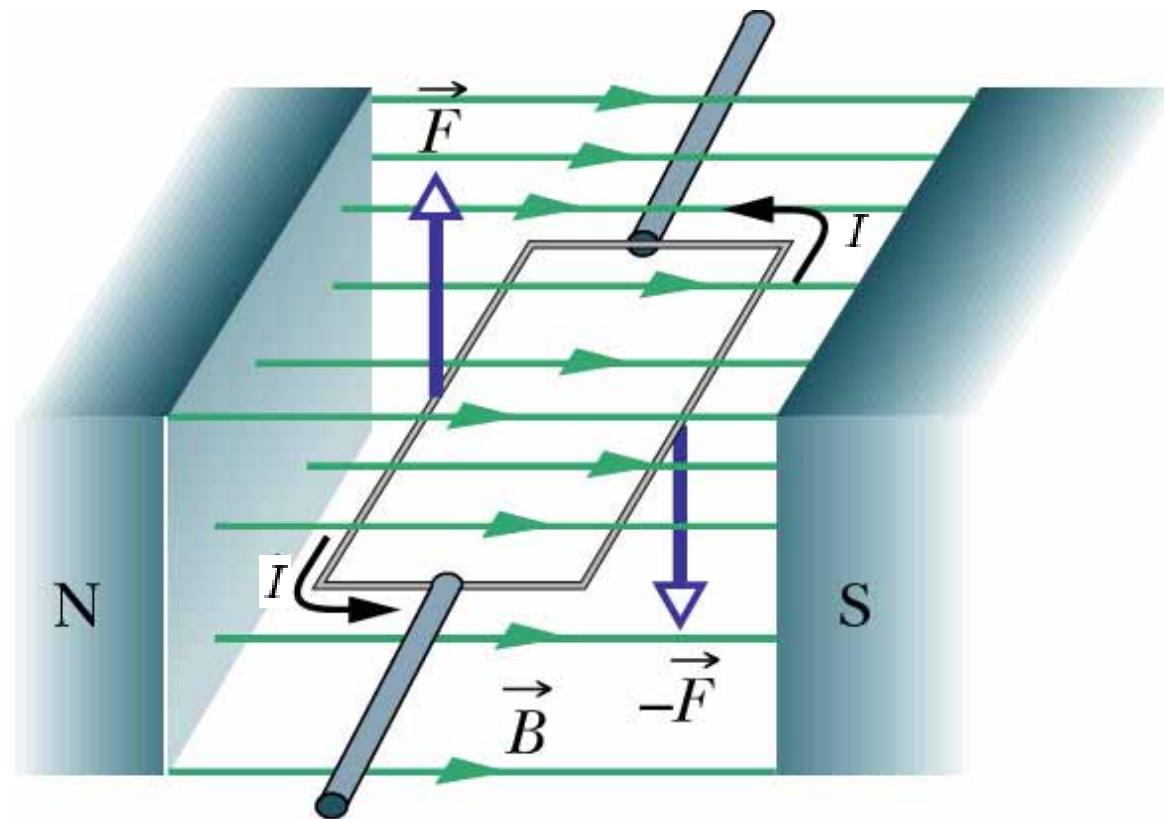
Demtröder – Experimentalphysik 2 (2013), Abb. 4.4

## zu 4.6: Generator



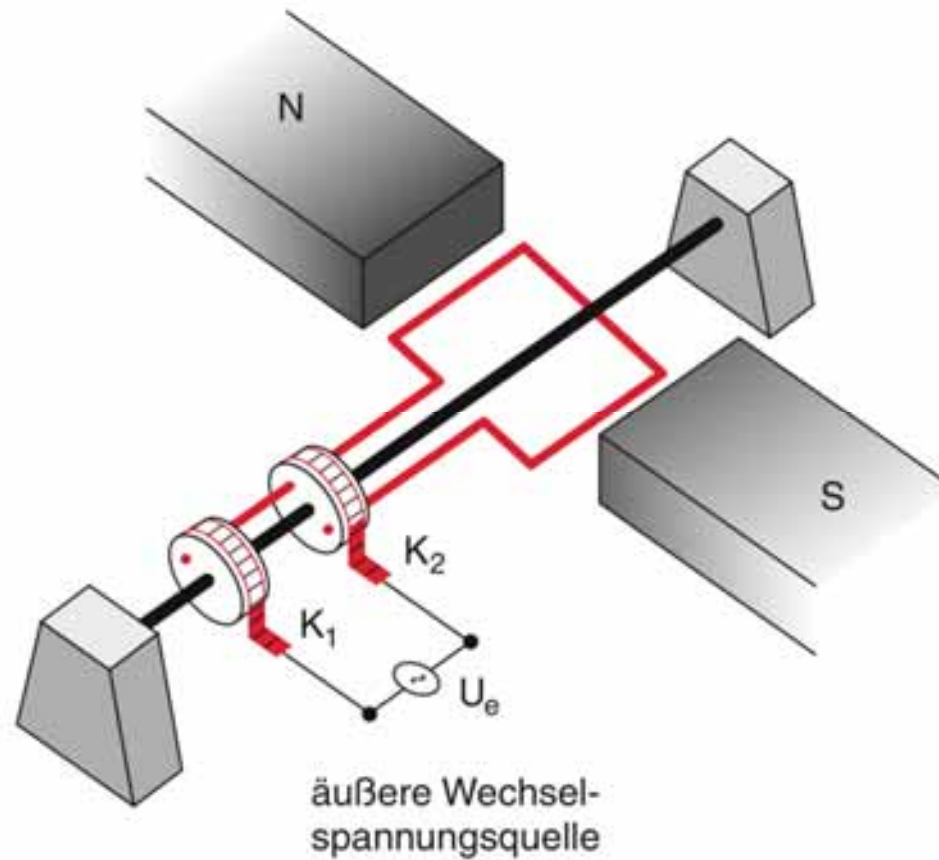
Demtröder – Experimentalphysik 2 (2013), Abb. 4.5

## zu 4.2: Leiterschleife im Magnetfeld



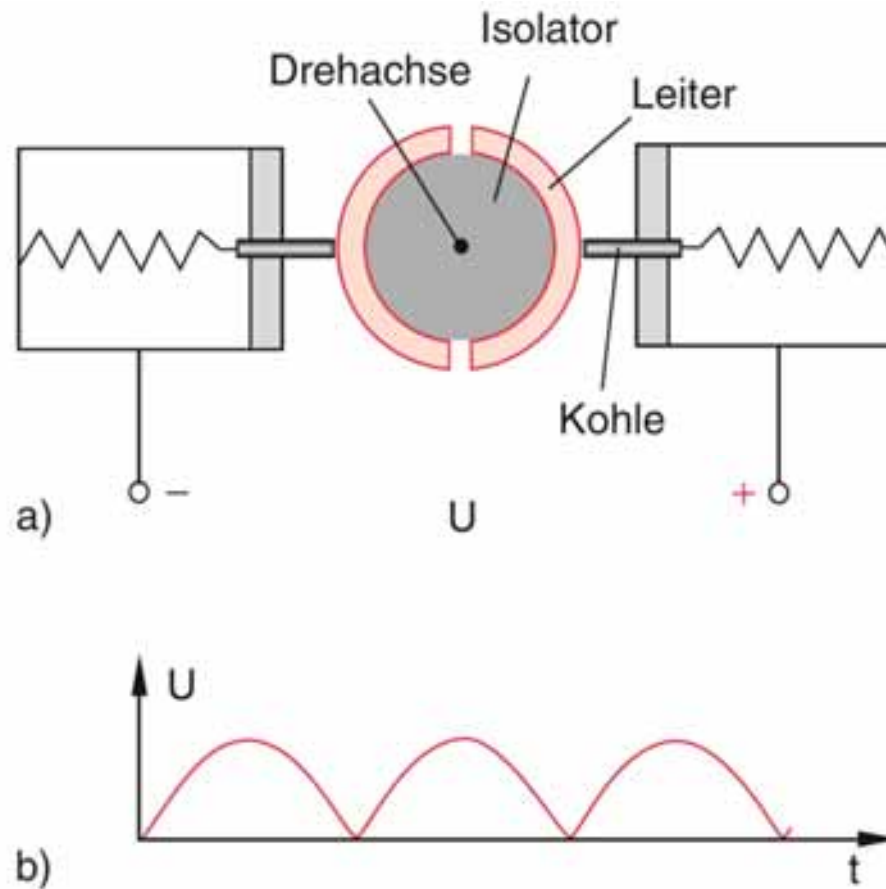
Halliday-Resnick-Walker (2003) Abb. 29.20

## zu 4.6: Wechselstromsynchronmotor



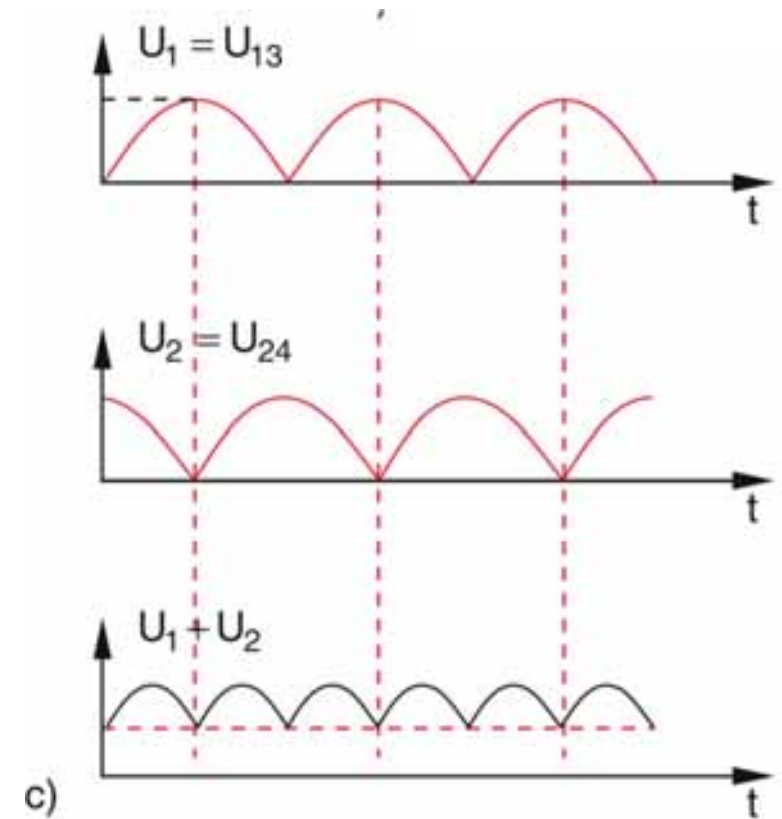
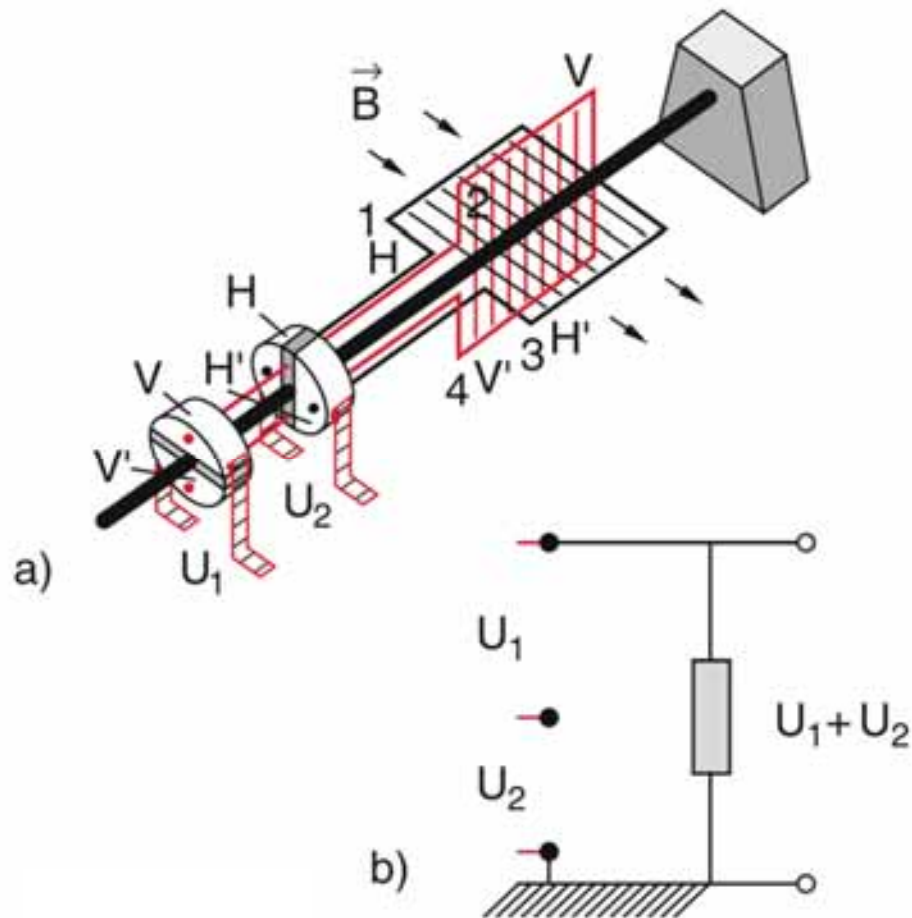
Demtröder – Experimentalphysik 2 (2013), Abb. 5.1

## zu 4.6: Gleichstromgenerator bzw. -motor



Demtröder – Experimentalphysik 2 (2013), Abb. 5.2

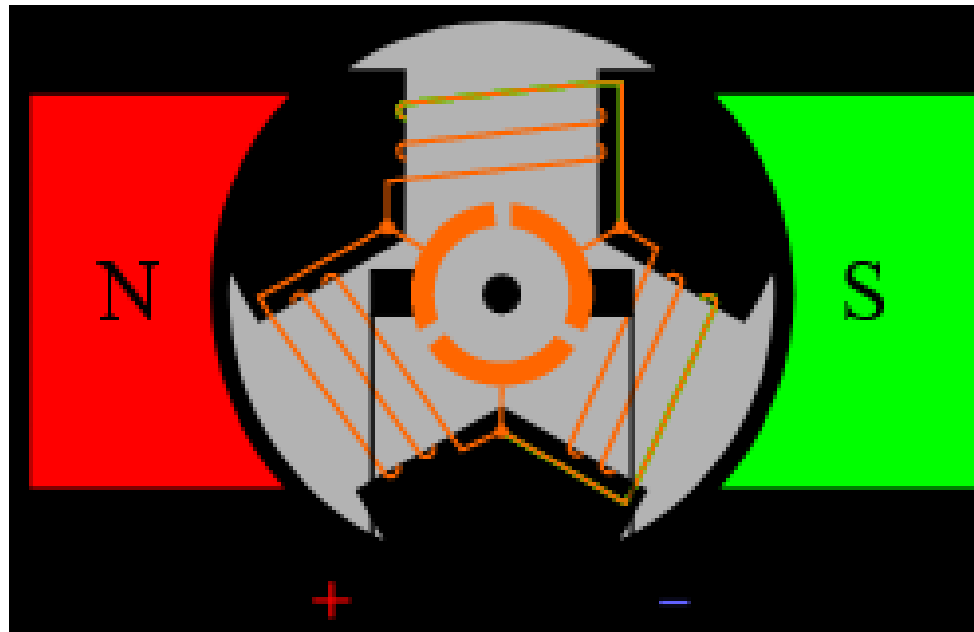
## zu 4.6: Gleichstromgenerator bzw. -motor



Demtröder – Experimentalphysik 2 (2013), Abb. 5.2

## zu 4.6: Gleichstromgenerator bzw. -motor

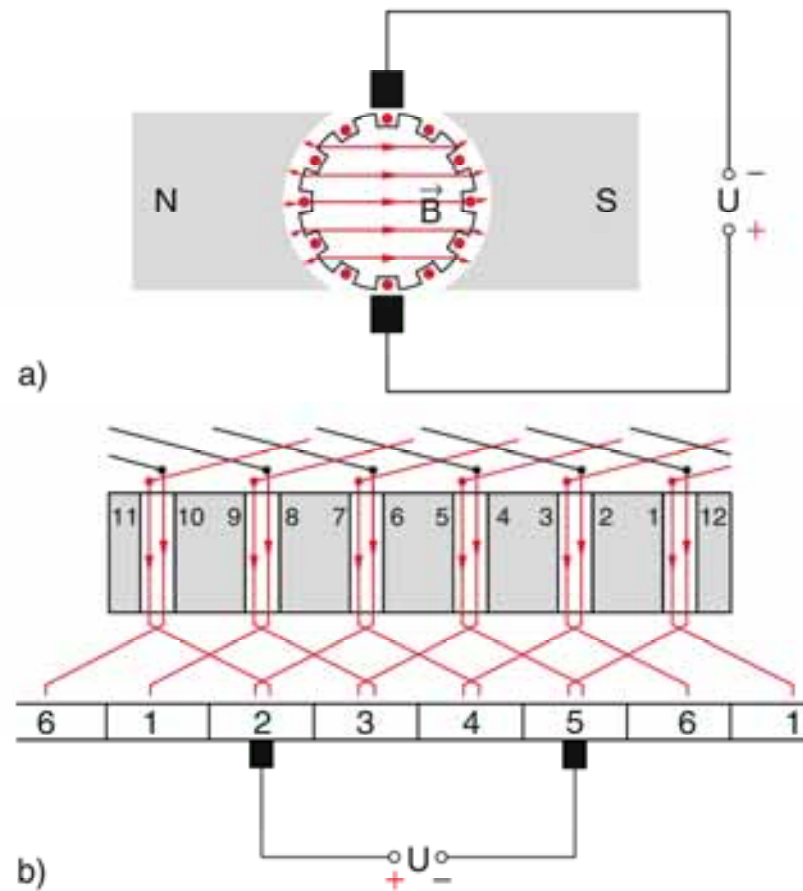
Trommelanker  
mit magnetisierbarem  
Eisenkern



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## zu 4.6: Gleichstromgenerator bzw. -motor

Trommelanker  
mit magnetisierbarem  
Eisenkern

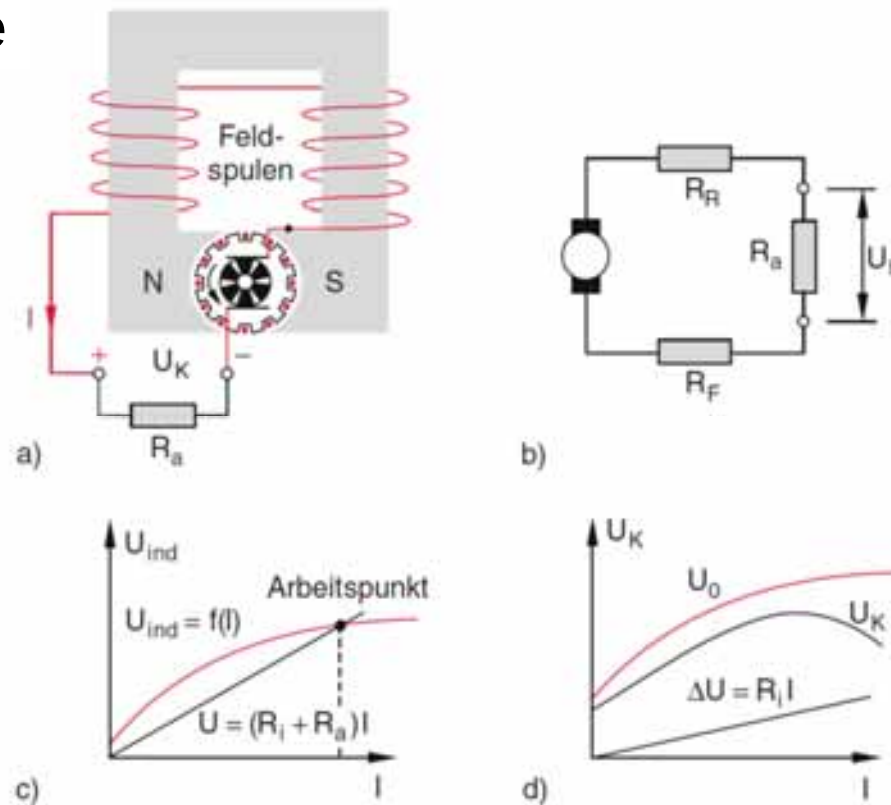


Demtröder – Experimentalphysik 2 (2013), Abb. 5.5



# zu 4.6: Gleichstromgenerator bzw. -motor

## Hauptschlussmaschine



Demtröder – Experimentalphysik 2 (2013), Abb. 5.7